

CSN5 INFN-LNF

A. GIRIBONO COORDINATRICE LNF

CDL PREVENTIVI 6 LUGLIO 2026

CSN5 update

6 Detector R&D

3 Interdisciplinary

Bando Call
(June 8th 2026)

Next CSN5 meeting

14-17 July 2026

Future CSN5 meeting

7- 11 September 2026

'Standard Exp.'
(July 4th 2026)

12 Detector R&D

9 Interdisciplinary

5 Accelerator R&D

Bando Grant Giovani

Deadline 31st July 2026 at 11.59 am

A. Quaranta

Grant Giovani

- Nuovo bando in uscita primi di luglio con scadenza 31 luglio.
- 5 anni dal dottorato.
- Ammessi Tecnologi PNRR e RTDA (secondo selezione GE).
- Il Panel decreta ammesso/non ammesso con giudizio analitico (specie per i non ammessi).
- Si seleziona a priori il «Responsabile Scientifico» all'interno della struttura. La definizione è obbligatoria sebbene il ruolo non sia quello di un tutor scientifico ma piuttosto di un referente per la commissione per comunicazioni, criticità, problematiche (è tenuto a comunicarle tempestivamente).
- Obbligatoria la lettera di approvazione da parte del Direttore della struttura.
- Max 75k il primo anno e 75k il secondo (si può sforare per piccole cifre).
- Tutto e solo in inglese.
- Punteggio finale 16+4.

Bando e template al link https://web.infn.it/csn5/images/PDFDocuments/28978.pdf

CSN5-LNF overview 2026

People= 77 (+13 %) : **Ric. 36** (+16%), **Tec. 41** (+11 %), **FTE= 27.7** (+30 %)

DETECTORS 2

- TANGO (RL)
- MSPC **(Grant Giovani)**

INTERDISCIPLINARY 2

- QUARTET (RL)
- SPHINX *

ACCELERATORS 9

- ASTERIX
- BOND (call) (RL)
- CROWN *(RL)
- H2BTF * (call) (RL)
- SL_BETATEST
- SUPERMAD (RL)
- ALAN
- HARMONIQ
- GIOTTO (RL)

* CLOSING

New proposals for 2027

New proposals con RN @ LNF

1. ARGUS — RN A. Michelotti
2. REDD — RN A. De Santis

New proposal GRANT giovani:

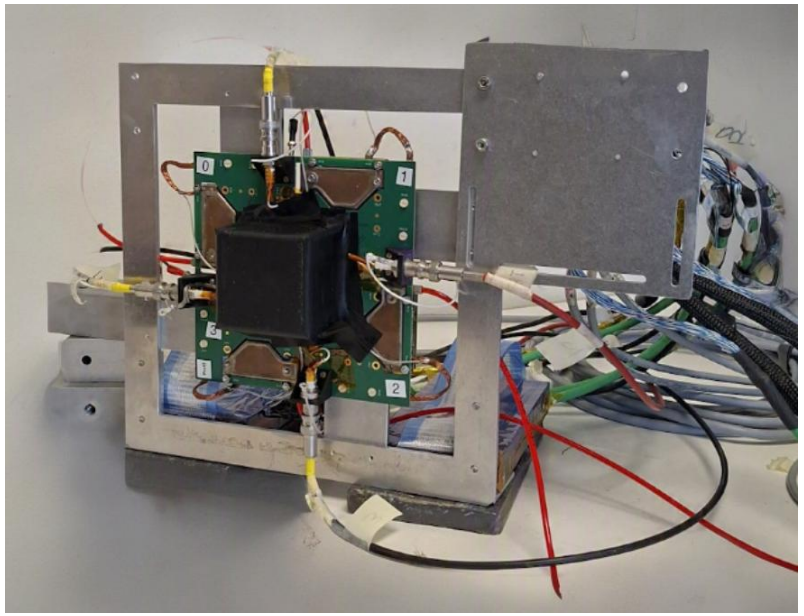
1. TBA

New proposals con RL @ LNF

1. AETHER — RL E. Diociaiuti
2. Piezo-POP — RL L. Faillace
3. SILVER — RL S. Dell'Agnello
4. CRIMSON (CALL) — RL I. Sarra
5. CRYOMIR (CALL) — RL M. Angelucci
6. nkeep (CALL) — RL R. Bedogni

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Proposta di Esperimento in CSN5 (INFN)



AETHER

Advanced Electronics for Tomography and High Energy Readout

3 anni (2027-2029)

Units: INFN Fe, INFN Bologna, LNF &Star @ UniCal, INFN Torino

Research line: Detector, Computation & Electronics

G. Cibinetto (Resp. Naz.) – INFN FE

Eleonora Diociaiuti (Resp. Loc.) – LNF

FTE 0.2 @LNF (0.1 E. Diociaiuti, 0.1 B. Buonomo)

FTE tot 0.2

The main goal of this experiment is to enhance and validate a CZT-based γ-spectroscopy readout platform with improved sensitivity and dynamic range, targeting applications in medical imaging, space astrophysics, and accelerator diagnostics.

Proposed Workflow of the Project

WP1: Electronics Enhancement

- New adapter card to minimize the noise
- Design and production of analog ASIC preamplifier
- Development of preamplifier mezzanine
- Enhanced system lab characterization (source/LARIX)

WP2: Space applications

- Sensor response simulation
- Sensor procurement and characterization
- Test with new adapter card
- Full enhanced system test with polarized photons



WP3: Accelerator applications

- Sensor response simulation
- Sensor procurement and characterization
- Test with new adapter card (BTF/STAR)
- Full enhanced system test (BTF/STAR)

WP4: Medical applications

- Test with new adapter card
- Full enhanced system test at LARIX/Hospital with SPECT and PET photons

Contributors connected to WP3

Skills & Facilities: Beam diagnostic, simulation, BTF, STAR @ UniCal

LNF (0.2 FTE)

- Bruno Buonomo – BTF Technical Manager
- Eleonora Diociaiuti – BTF Staff

UniCal *

- Vincenzo Formoso – Head of the µTomo2 and SoftX Beamlines
- Tommaso Caruso – Head of the Advanced Materials Spectroscopy Laboratory
- Maria Caterina Crocco – Postdoctoral Researcher, µTomo2 and SoftX Beamlines
- Antinella Nicolino – PhD Student
- Andrea Solano – Electronics and Control Technician, STAR

***University staff/students and will not be affiliated with INFN.**

Financial request

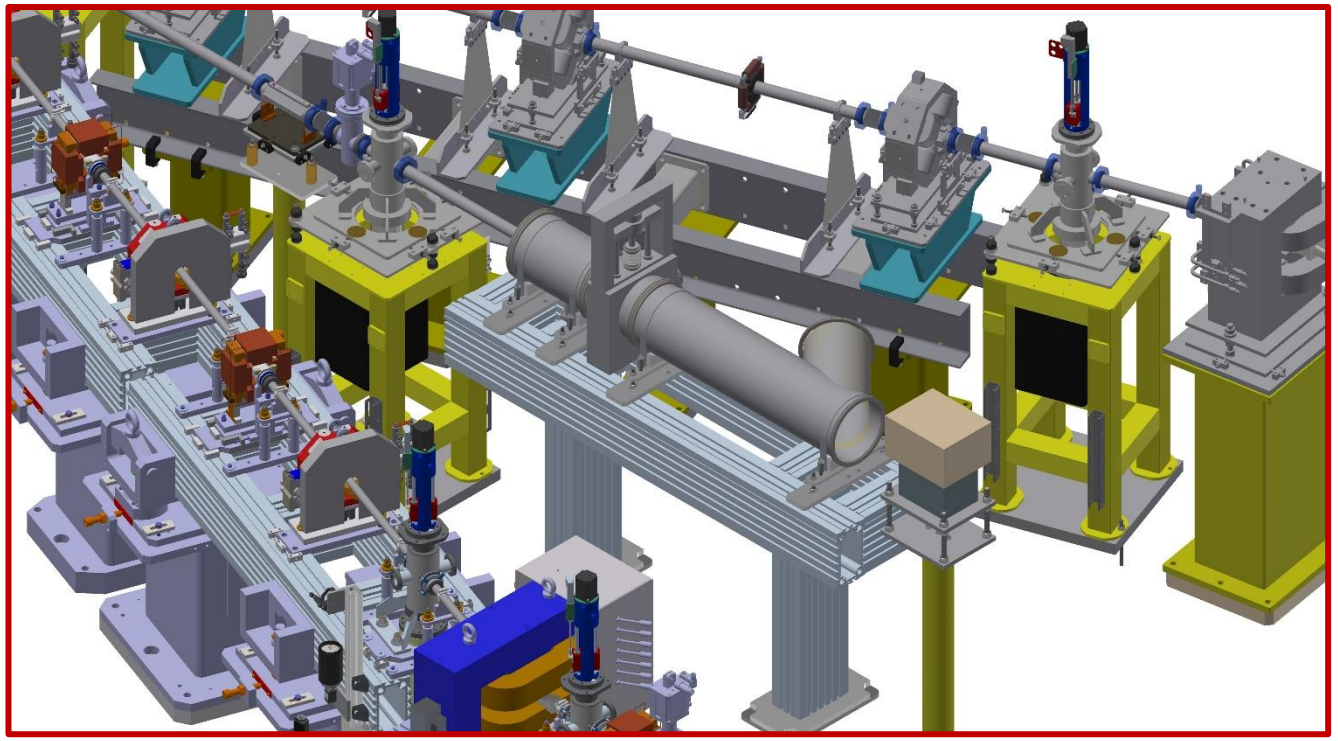
For the initial LNF request:

- Travel funds: 2.5k€/year
- 1 k€ consumables

- Before purchasing a dedicated CZT sensor for BTF activities, preliminary tests with sensors already available within the collaboration would be highly valuable. These measurements would help confirm the usefulness of these detectors for the BTF application and guide a possible future procurement on a more solid experimental basis.

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Proposta di Esperimento in CSN5 (INFN)



PiezoPOP

Piezoelectricity Proof of Principle

3 anni (2027-2029)

Units: MI,LNF,FE,RO1

Research line: Accelerators

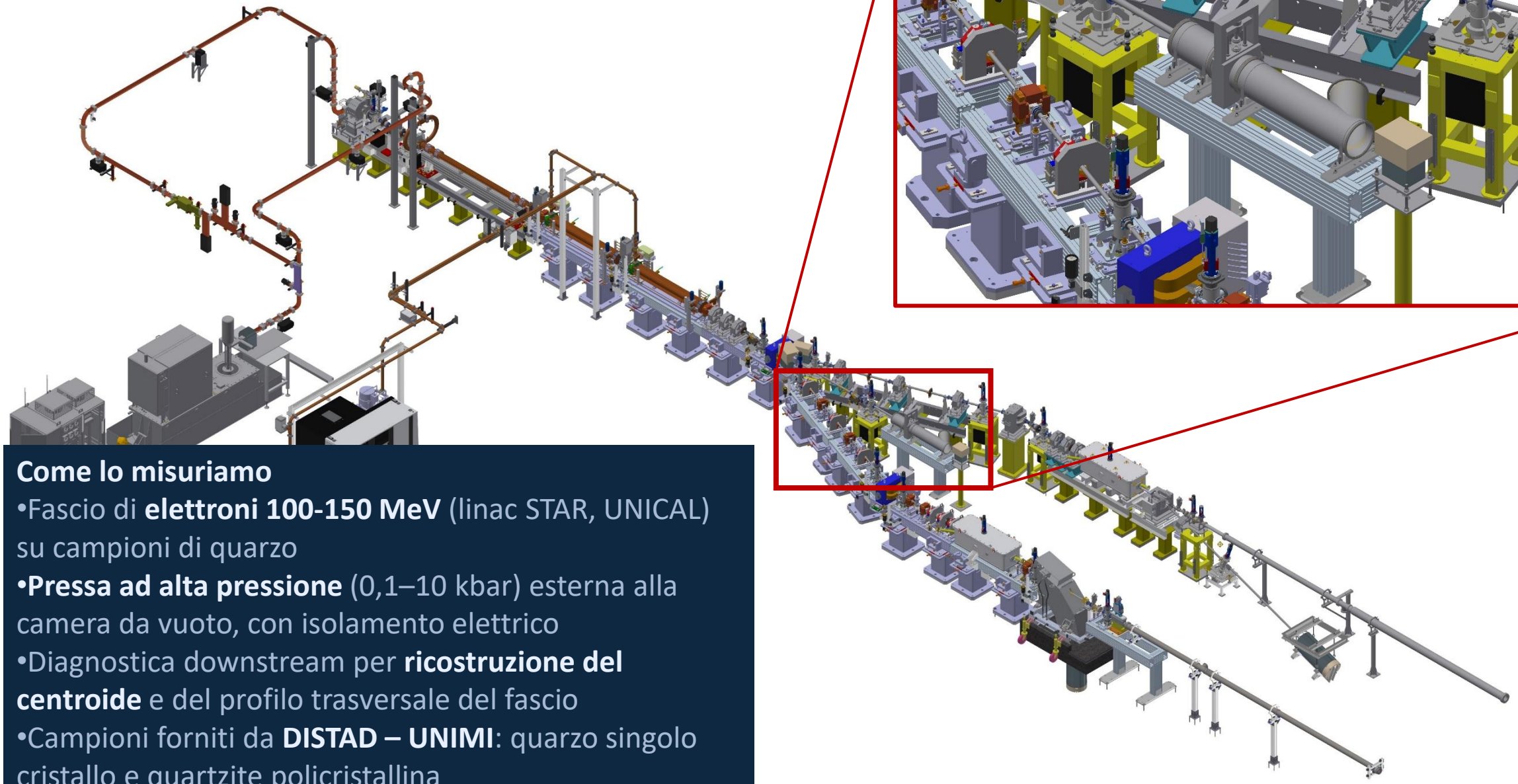
M. Rossetti Conti - (Resp. Naz.)

L. Faillace - (Resp. Loc.)

FTE <u>1.15@LNF</u>

The main goal of the experiment is to experimentally characterize, for the first time, the coupling between piezoelectric fields in quartz under mechanical stress and the dynamics of charged particles propagating through it (100-150 MeV electrons produced at STAR).

PiezoPOP @ STAR



Come lo misuriamo

- Fascio di **elettroni 100-150 MeV** (linac STAR, UNICAL) su campioni di quarzo
- **Pressa ad alta pressione** (0,1–10 kbar) esterna alla camera da vuoto, con isolamento elettrico
- Diagnostica downstream per **ricostruzione del centroide** e del profilo trasversale del fascio
- Campioni forniti da **DISTAD – UNIMI**: quarzo singolo cristallo e quartzite policristallina

WPs e Tasks

WP0 — Project Coordination and Integration
Leader: M. Rossetti Conti (MI)
- Overall coordination and integration of project activities.

WP1 — Samples, Press System, and Characterization
Leader: M. Voltolini (MI)
- Design and fabrication of the mineral press system.
- Selection, preparation, and conditioning of quartz and polycrystalline samples.
- X-ray diffraction analysis of the samples.

WP2 — Interaction Chamber and Beamline
Leaders: **L. Faillace (LNF)** e M. Rossetti Conti (MI)
- Procurement of beamline hardware, including magnets, flanges, pipes, viewports, pumps, diagnostics, and optical components.
- Outsourced installation of beamline sections, the press system, diagnostics, and the interaction chamber.
- Beamline alignment and integration.

WP3 — Development of Piezoelectric Simulation Framework
Leaders: I. Drebot (MI) e G. Paternò (FE)
- Implementation of piezoelectric fields in GEANT4.
- Production of field maps with FEM code (COMSOL).
- Integration of field maps into external simulation tools.
- Modelizations of quartz and novaculite geometries.

WP4 — Beam Test and Diagnostics
Leaders: **(LNF)** e (Roma1)
- Detector studies.
- Simulation of the new beamline with CI.
- Measurement of electron deflection.
- Signal validation.
- Elasto-optic measurements.

WP5 — Theoretical Extensions
Leader: C. Galelli (MI)
- Extension to atmospheric muons.
- Scaling to broader geophysical scenarios.

Participants and Financial Request@LNF

- **Minimal commitment from INFN-LNF**, with **no impact on LNF technical services**.

Year 1 (2027) and 2: provide **scientific support only**, including:

- Beam dynamics simulations.
- Design support for the vacuum beamline and detector required for the STAR experiment planned in Year 3.

- **No responsibility for procurement or installation**: All purchasing, integration, and installation activities will be handled by the **INFN Milan group**.

- **Experimental phase (Year 3):**

- LNF will provide support through a limited number of **missions to STAR**.
- The LNF team will assist with machine operation and beam tuning, as it is currently **the only group capable of operating STAR and delivering the required electron beam conditions**.

LNF	FTE
L.Faillace (Resp. Loc.)	0.2
S. Bini	0.15
A. Curcio	0.3
E. Chiadroni	0.2
A. Giribono	0.1
E. Tassi (Cosenza)	TBD
P. Pagliusi (Cosenza)	TBD
Tot./Unit	**>1.15**

	Descrizione	Richieste (k€)
missioni	Machine time at STAR in order to measure the electron beam parameters needed for this experiment. 2x running weeks for 3 people	4
Totale		**4**

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Proposta di nuovo Esperimento INFN-CSN5

SILVER

(**S**eismic **I**solator for **L**unar gra**V**itational-wave d**E**tector with **R**etroreflectors)

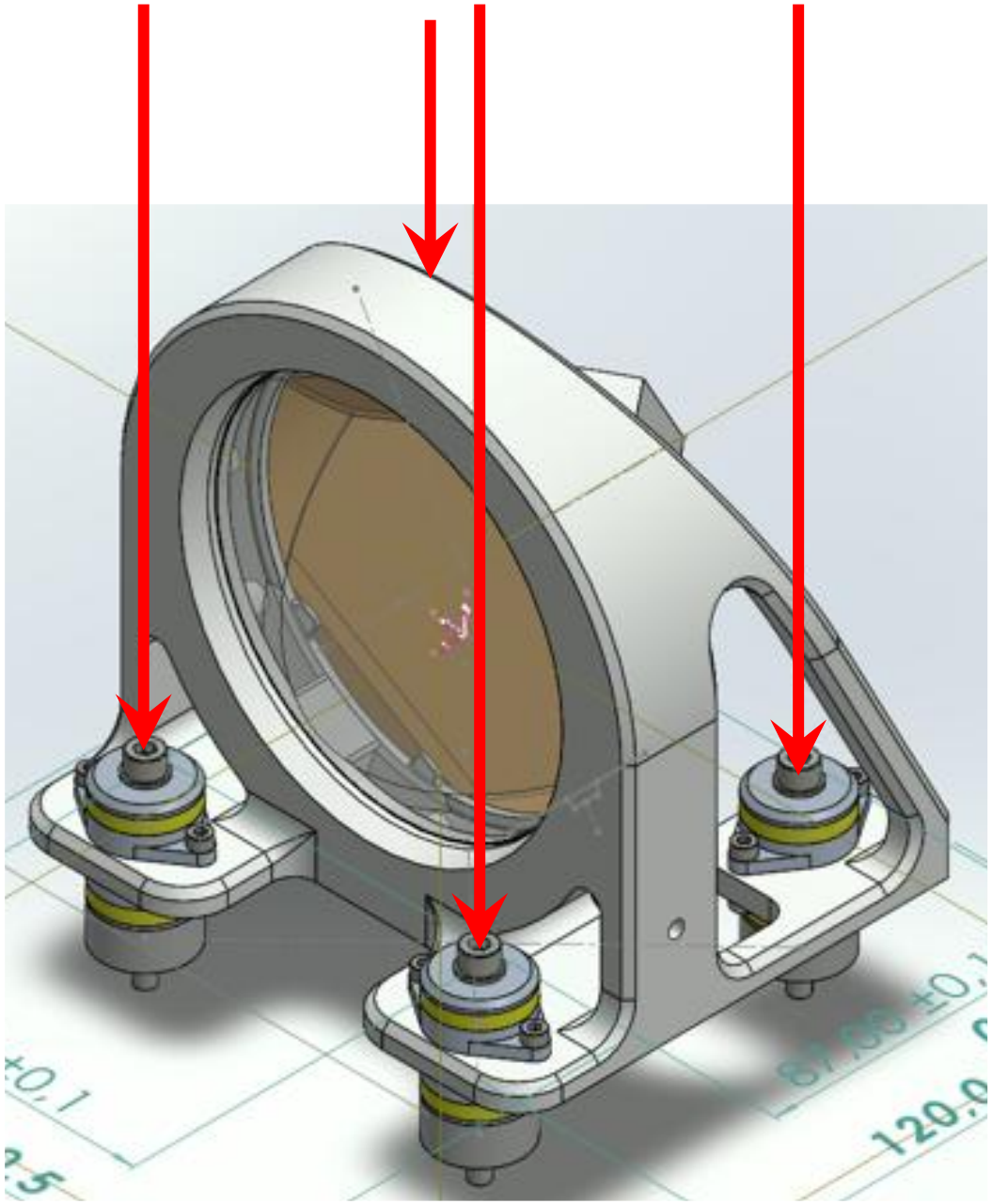
3 years (2027-2029)
Units: INFN-LNF (RL, SCF_Lab),
INFN-Pisa (RN, Virgo Group)
Research line: RIVELATORI/ELETTRONICA

RL LNF: S. Dell'Agnello

FTE 1.5@LNF **FTE tot 4.5**

The main goal of this experiment is the design, construction and test of an **innovative prototype of a seismic isolation** device (INFN-Pisa) equipped with a **lunar laser retroreflector customized for laser interferometry (INFN-LNF, SCF_Lab Research Group)** to be implemented in a lunar Gravitational-Wave (GW) observatory operating in the deciHertz band (0.1 - 10 Hz) on the surface of the Moon

Seismic isolation TDB, imagined from above applied at 4 interface screws)



Milestones

WP1: Design and simulations

- Milestone 1: Full simulation of the prototype completed and validated
- Milestone 2: Noise budget completed
- Milestone 3: Release of engineering drawings

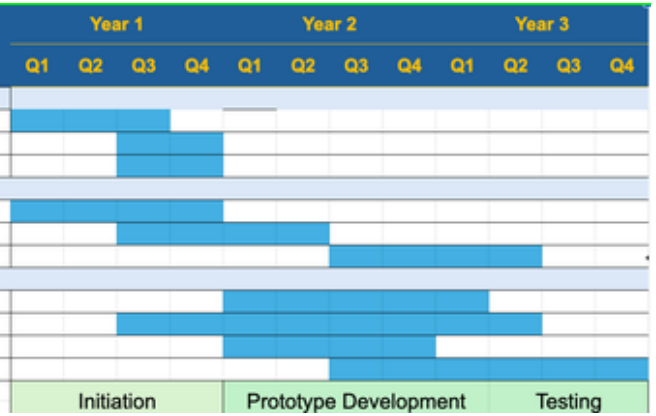
WP2: Retroreflector-based optics (LNF)

- Milestone 1: Model of Sun shade for retroreflector
- Milestone 2: Simulation of optical layout with reflector
- Milestone 3: Characterization of retroreflector finalized
- Milestone 4: Reflector integrated in prototype

WP3: Seismic isolator prototype

- Milestone 1: Procurement completed
- Milestone 2: Assembly of mechanics completed
- Milestone 3: LVDTs & actuators assembled
- Milestone 4: Report on evaluation of optical sensore applied to seismic isolation
- Milestone 5: Control method & SW completed
- Milestone 6: Report on comparison of AI for control completed
- Milestone 7: Vibe tests on components completed
- Milestone 8: Thermal tests completed

WP	Task	Leading Unit	Start Month	End Month	Duration (Months)
1	**Design and Simulations**				
	T1.1 Mechanical Simulations	PI	0	9	9
	T1.2 Noise budget	PI	6	12	6
	T1.3 Prototype Drawings	PI	6	12	6
2	**Retroreflector-based optics**				
	T2.1 Optical Simulations and design customization	LNF	0	12	12
	T2.2 Characterization of customized reflector	LNF	6	18	12
	T2.3 Interface and integration in the prototype	LNF	18	30	12
3	**Seismic Isolator prototype**				
	T3.1 Building of mechanical components	PI	12	27	15
	T3.2 Sensors and actuators	PI	6	30	24
	T3.3 Control infrastructure	PI	12	24	12
	T3.4 Full payload Test and Characterization	LNF	18	36	18



WP	Milestone	Task	Month	Description
1	M1.1.1	T1.1	9	Full simulation of the prototype completed and validated
	M1.2.1	T1.2	9	Noise budget completed
	M1.3.1	T1.3	12	Release of the engineering drawings
2	M2.1.1	T2.1	9	Model of shade for the retroreflector
	M2.1.2		12	Simulation of the optical layout using reflectors
	M2.2.1	T2.2	18	Characterization of the retroreflector finalized
	M2.3.1	T2.3	30	Reflector integrated in the prototype
3	M3.1.1	T3.1	18	Procurement of all component completed
	M3.1.2		27	Assembly of mechanics completed
	M3.2.1	T3.2	15	LVDTs and actuators assembled
	M3.2.2		24	Report on evaluation of optical sensors applied to seismic isolation
	M3.3.1	T3.3	21	Control method and software completed
	M3.3.2		18	Report on comparison of AI for control completed
	M3.4.1	T3.4	30	Vibration tests on components completed
	M3.4.2		36	Thermal tests completed

Plan and Financial Request@LNF for 2027

	Descrizione	Richieste (k€)	Totale Rich. (k€)
Consumo	Mechanical mounts for the interchangeable fitting of existing retroreflectors (4, 3, 2 inches) and the 4-inch flat mirror for optical measurements. Geometric adaptors for fitting 4-, 3- and 2-inch retroreflectors	3.0	3.0
Missioni	Annual mini-workshop of the international collaboration LILA; joint laboratory set-up/testing and collaborative meetings in Pisa (2 ppl, 3 per year)	3.0 3.0	6.0
Inventario	4-inch flat mirror (accuracy wave/20) for comparing interferometric performance with existing laser retroreflectors of 4, 3, and 2 inches in diameter. As per the Edmund Optics online price + VAT	1.5	1.5
Totale			**10.5**

Preliminary Financial Request for 2028 and 2029:

- Year 2028: 20 kEuro
- Year 2029: 30 kEuro

LNF	FTE
S. Dell'Agnello (Resp. Loc.)	0.3
G. Bargiacchi	0.2
N. Burzillà	0.2
R. Campagnola	0.2
V. Cardillo M. Fioravanti	0.2 0.2
A. Epifanii	0.2
Tot./Unit	**1.5**

Collaborations

We are currently co-founders of the LILA (Lunar Interferometer Laser Antenna) collaboration, consisting of USA groups from LIGO led by Univ. of Vanderbilt (working on the laser system, DAQ system, their protection from radiation, thermal perturbation and dust, plus other system aspects), French group IPGP (working on seismometry for veto, noise evaluation and background subtraction). We (INFN) work on the far detector ststion (seismic isolation and retroreflector-based optics).

We made a response to ESA Call for Ideas 2025 (loaded online Preventivi) and in 2026 a proposal in the US, but funding will go to the US side.

Invited Talks related to SILVER

- 1 Pisa talk and 1LNF talk at every annual meeting of the discipline / community since 2021 (first workshop organized by S. Katsanevas at EGO-Pisa.

Publications

- White Paper of the LILA Collaboration 2025

CRIMSON, CRIlin–MPGD calorimetry with Smart OptimizatioN

RN: Luigi Longo
RL: Ivano Sarra

Goals:

- development of an **integrated electromagnetic (ECAL) and hadronic (HCAL) calorimeters system**:
  - ECAL:
    - semi-homogeneous PbF_2 calorimeter (CRILIN), energy resolution ~ 5%/√E, time resolution better than 40 ps
    - longitudinal dimensions ~ 1 λ_I → impact on HCAL
    - recover the hadronic shower energy not seen by CRILIN, from the transversal development of the hadronic shower
  - HCAL:
    - Micro-Pattern Gas Detector as active layers for HCAL (MPGD-HCAL)
    - Improve MPGD timing resolution to reduce the contribution due to delayed neutrons and affecting the shower energy reconstruction
  - reconstruction algorithm based on the shower development topology in both ECAL and HCAL → use of Dynamic Graph Convolutional Neural Networks (DGCNN)
  - construction of a CRIMSON prototype combining CRILIN and MPGD-HCAL prototypes:
    - unique DAQ system based on VMM3a electronics
    - validation of the DGCNN reconstruction algorithm at test beam
- **MPGD-HCAL as muon tracking system:**
  - evaluate the use of capacitive-sharing MPGDs in the last HCAL layers to perform muon track seed reconstruction, before performing a full muon track reconstruction towards the IP → FCC-ee CLD detector will be considered as benchmark
- **Industrial production of MPGD:**
  - collaboration with ELTOS S.p.a. for industrial production of a large area Micromegas (100x50 cm^2) → a breakthrough towards detector MPGD industrial production for the next experiments
- Condensate all the above points in a proposal for a **large-scale CRIMSON demonstrator**

Progetti sinergici: DRD1, DRDCalo, RD_MuCol, RD_FCC

CRILIN Project Status 2026

FINAL PROTOTYPE (2 M_R, 22 X_0):

5 layers, 7x7 crystals, ~ 250 channels

- PbF_2 1.3x1.3 cm^2 with 0.1mm tolerance
- 4 SiPMs 3x3 mm^2 per crystal

Improved readout/mechanics:

- Kapton strip for SiPM connection
- New custom FEE → partially ready
- 8 CAEN V1742 digitizer
- Aluminum alveolar matrices with:
  - 150 um septa,
  - 2mm external stackable envelope
  →production completed
- ***Assembly designed to be:***
  - ***experiment-ready***
  - ***radiation-hard***

Involved groups:

INFN: LNF, Roma1, Padova, Torino, Firenze
Enea Casaccia, HZDR (Dresden)



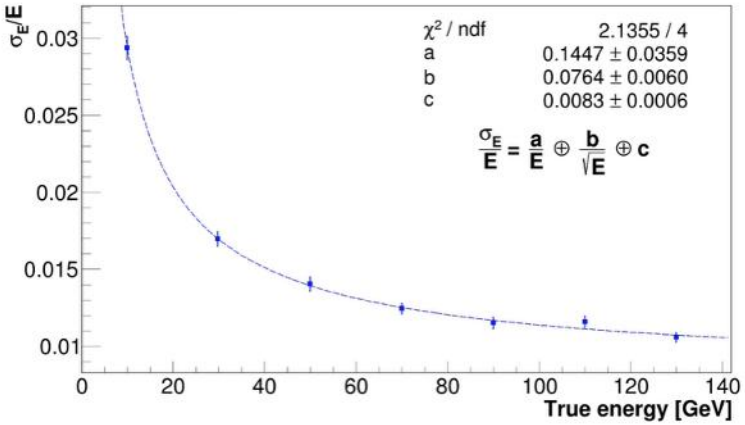
Test Beam @ CERN SPS (energy resolution) end of June 2026

Test Beam @ CERN PS with MPGD HCAL end of July 2026



Simulation: Energy Resolution ~ 7.5%/√E

- 0.2 p.e./MeV per crystal (conservative w.r.t. TBs)
- gaussian noise σ = 10 MeV
- 30 MeV energy threshold per crystal



CRIMSON: activities in which LNF's group is involved

2027:

- **WP2 - Simulation:**
  - Geant4 simulation of CRILIN and MPGD-HCAL calorimeters.
  - Implementation of CRILIN and MPGD-HCAL in CLD simulation.
- **WP3 - CRILIN and MPGD-HCAL integration:**
  - Development of the reconstruction algorithm for the hadronic energy deposit in CRILIN, parametrized from the shower RMS, with optimization of CRILIN layers length.
  - Development of common DAQ, based on VMM3a readout.

2028:

- **WP3 - CRILIN and MPGD-HCAL integration**
  - Development of ML-based energy reconstruction using DGCNN architectures, based on the info provided by CRILIN and MPGD-HCAL calorimeters, and it evaluation on CRIMSON prototype.
  - Commissioning of CRIMSON prototype, made of CRILIN and MPGD-HCAL prototypes.
  - CERN PS test-beam campaign for ML-reconstruction model validation.

2029:

- **WP3 - CRILIN and MPGD-HCAL integration**
  - CERN PS test-beam campaign for ML-reconstruction model validation.
- **WP6 - CRIMSON detector concept and publications**
  - Definition of requirements for future electronics and DAQ.
  - Evaluation of TRL increase.
  - Preparation of European-scale proposal for large CRIMSON demonstrator.

CRIMSON: Anagrafica e Richieste

Sedi partecipanti:

Bari, Firenze, LNF, Napoli, Roma3, Torino

RN: Luigi Longo
RL: Ivano Sarra

Richieste finanziarie in kE:

- ~70k euro su tutto il progetto per i 3 anni
- 2 Incarichi di Ricerca fascia 1 (2 anni/each)

Richieste Servizi (totali nei 3 anni):

- Electronics and automation (4MP)
- Detector development (4MP)

FTE @LNF 1.6 + 2 postdoc

	FTE		
Personale	**2027**	**2028**	**2029**
Ivano Sarra	30%	30%	30%
Fabio Happacher	20%	20%	20%
Stefano Miscetti	10%	10%	10%
Simone Bini	20%	20%	20%
Claudio Cantone	20%	20%	20%
Vittoria Ludovica Ciccarella	20%	20%	
Alessandro Russo	30%	30%	30%
Sergio Ceravolo	30%	30%	30%
PostDOC LNF-1	50%	100%	50%
postDOC LNF-2	50%	100%	50%

Proposta di Esperimento in CSN5 (INFN)

Istituto Nazionale di Fisica Nucleare

CRYOMIR

(Cryogenic Low Noise Mirrors)

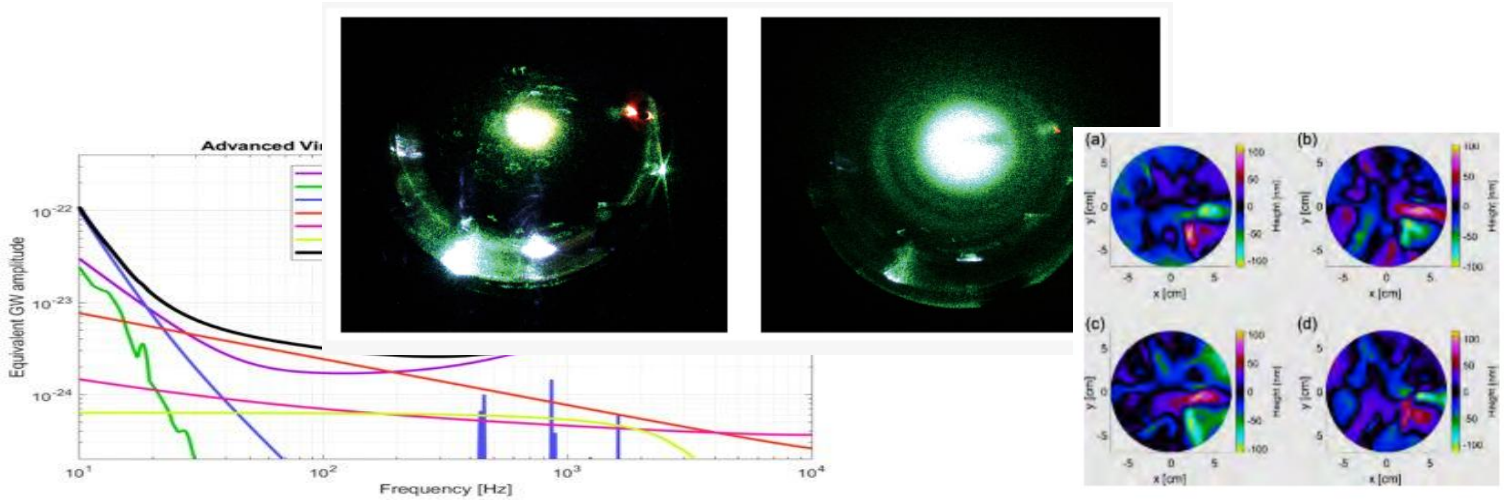
3 anni (2027-2029)
Units: Tor Vergata, LNF,Ferrara, Firenze, Genova, Milano, Padova, Salerno
Research line: GW detectors, Cryogenics, Optical Metrology

Elisabetta Cesarini (RM2, RN)

Roberto Cimino (RL-LNF)

FTE 1.1 @LNF

FTE tot 9.4



CRYOMIR is a project aimed at understanding and validating the optical, mechanical, and surface phenomena that affect the performance of **multilayer optical mirrors at cryogenic temperatures**. Currently, no cryogenic mirror architecture has been fully qualified, mainly due to the incomplete understanding of key factors such as optical absorption, mechanical losses, thermo-mechanical stress, and surface instabilities.

To address this limitation, the project develops an integrated cryogenic metrology approach capable of performing reproducible and cross-correlated measurements on identical samples under controlled conditions. In this context, **INFN facilities will be upgraded and standardized procedures will be introduced, combining multiple optical and mechanical characterization techniques, along with studies of contamination and frosting effects.**

In parallel, realistic **multilayer coatings will be designed and optimized for low-temperature operation**. The final outcome will be the validation of a cryogenic mirror operating reliably under controlled conditions, providing key technologies **for next-generation interferometers** and establishing a reusable platform for precision optical metrology.

Proposta di Esperimento in CSN5 (INFN)

Istituto Nazionale di Fisica Nucleare

CRYOMIR

(Cryogenic Low Noise Mirrors)

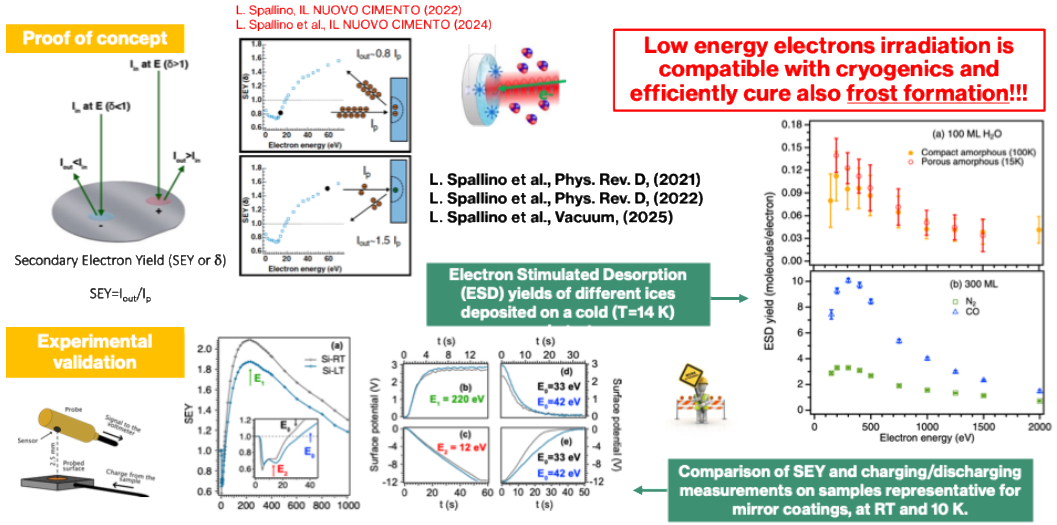
3 anni (2027-2029)
Units: Tor Vergata, LNF,Ferrara, Firenze, Genova, Milano, Padova, Salerno
Research line: GW detectors, Cryogenics, Optical Metrology

Elisabetta Cesarini (RM2, RN)

Roberto Cimino (RL-LNF)

FTE 1.1 @LNF

FTE tot 9.4



Background: Proof of concept and experimental validation studies of possible use of electrons irradiation for the frost and charging mitigation.

LNF objective and activities 2027-2029 (WP5)

Objective: Study and quantify **surface-induced effects limiting optical stability in cryogenic environments** to implement mitigation strategies for frost formation (Electron Stimulated Desorption [ESD] and/or CO2 laser irradiation) and surface charging (low energy electron irradiation) compliant with cryogenics.

Activities: ESD and charging measurements; support the design and setup of a dedicated cryogenic optical cavity for surface stability test and provide guidance on the implementation of cryogenic systems across different laboratories.

Plan and Financial Request@LNF for 2027

Activities (2027)

WP5.1 Electron irradiation dynamics:

- Electron irradiation dynamics for frost mitigation and surface charge neutralization at RT on small optical samples.

WP5.2 Design and development of a multi-technic optical cavity (in collaboration with RM2):

- Design of multi-technique cryogenic optical cavity for irradiation and spectroscopic characterization.

	Descrizione	Richieste (k€)	Totale Richieste (k€)
Consumo	Componenti da vuoto	5	20
	Sensori per Criogenia	5	
	Lavorazioni Meccaniche	10	
Missioni	Meeting di progetto	3	3
Inventario	Controller temperatura	5	5
Totale			**28**

LNF	FTE
R. Cimino (RL)	0.5
M. Angelucci	0.3
L. Spallino	0.2
R. Larciprete	0.1
Tot./Unit	**1.1**

nkeep – RN M. Marafini (Roma1)
Call CSN 5 – 2027-2029

Istituto Nazionale di Fisica Nucleare
Laboratori Nazionali di Frascati

Neutrons Key Element for Enabling Patient-tailored planning in particle therapy

Scientific Case

≈ 20 M new cancers/y diagnosed worldwide. 50% RT candidates

- RT-induced 2nd cancers due **Peripheral doses** (PDs) arise in 1% (10^5 / y)
- PDs are NOT included in TPS – need specific evaluations.
  - For photon/electron therapy, PDs and 2nd cancer risks were widely studied by Uni-Sevilla and cow.[1]
  - For hadron therapy (n-PDs) there are computational studies and very few experimental ones.
    - ❖ Experimental methods are passive, time-consuming, anisotropic and limited to 20 MeV
    - ❖ Simulation codes are not established/harmonised enough to be used for predictions
- Enhanced concern for young and pregnant patients
- Clinical-driven need exists for:
  - ❖ Better understanding the source term (energy/direction production)
  - ❖ Assessing PDs and creating established datasets to be used in the clinics
- **Neutrons could be used as monitor signal for online verification of the treatment**
- Neutrons should be **embedded in TPS** – greatly benefiting proton/carbon therapy or BNCT



DOI 10.3389/978-2-83251-093-3

[1] Book: **Out-of-field second primary cancer induction: Radiation-biology, epidemiology, dosimetry and modelling**.

nkeep
Call CSN 5 – 2027-2029

Istituto Nazionale di Fisica Nucleare
Laboratori Nazionali di Frascati

Target - *Particle therapy*

Deeper knowledge of n production and released doses + experimental validation for:

- Exploiting neutrons in treatment planning optimisation, delivery and verification
- Improve BNCT planning, currently suffering from limited knowledge of in-patient neutron spectra
- quantify their detrimental effects on patient health and provide PD datasets for clinical use
- Implementation in Monte Carlo tools for therapy planning (HT + BNCT) and clinical PD databases

Innovative aspects

- **Measuring energy, angle n production term using**
  - ❖ TOF + particle discrimination (Si, plastic, $CeBr_3$ array)
  - ❖ Optical fibers 3D tracker
- **Measuring PDs with techniques that are active, isotropic and complete in energy** :
  - ❖ Isotropically moderated SiC + ^6LiF
  - ❖ Indium and Bismuth activation spheres + CZT gamma spectrometer
  - ❖ Silicon detector coated with scattering radiators or Bismuth foils
- Characterisation of BNCT fields with micro-dosimeters, alanine dosimetry and n spectrometers.
- Model benchmarking in fast MC platforms (FRED) and TPS algorithms – providing updated fast MC and open TPS tools for improved planning of both BNCT and PT treatments

nkeep
Call CSN 5 – 2027-2029

Istituto Nazionale di Fisica Nucleare
Laboratori Nazionali di Frascati

Organisation

WP1	R. Bedogni LNF	PDs measurements and clinical datasets
WP2	E. Fiorina TO	n production term measurements (TOF + fiber tracker)
WP3	A. Schiavi RM1	fast MC and open TPS tools
WP4	I. Postuma PV	BNCT fields characterisation and related treatment planning developments

Requests

Anno	Consumo	Servizi	Inventariabile	Missioni	Totale
1	216,0	4,5	22,0	53,0	295,5
2	188,0	22,0	10,0	49,0	269,0
3	71,0	9,5	0,0	57,0	137,5

Sezione	Totale	Mesi contratto	Totale nKEEP
		2,75	
		dal 2 anno	
CT	13,00	12	
BO	59,00		
MI	104,00	24	
LNF	153,00	24	
LNS	11,00		
PV	73,00	18	
RM1	192,00		
RM3	9,00	12	
TIFPA	31,00		
TO	57,00	18	
Totale mesi		108	
Totale:	702,00	297	999,00

nkeep
Call CSN 5 – 2027-2029

Istituto Nazionale di Fisica Nucleare
Laboratori Nazionali di Frascati

Anagrafica nazionale

Sez	People	FTE
CT	6	1.5
BO	2	1
MI	12	3.7
LNF	8	2.8
LNS	5	1.7
PV	13	2.9
RM1	13	5.4
RM3	4	1
TIFPA	6	1.2
TO	9	2.8
TOT	**78**	**24 su 10 sezioni/Labs**

LNF Anagrafica 2027

- Roberto Bedogni 0.5
- Abner Ivan Castro-Campoy 0.2
- Claudio Cantone 0.2
- Tommaso Napolitano 0.1
- Matteo Beretta 0.1
- Elisa Minucci (ass) 0.2
- Antonino Pietropaolo (ass) 0.5
- Fabio Moro (ass) 0.5
- José Maria Gomez Ros (ass) 0.5

2.8 FTE

Richieste SPCM 2027

Piccole lavorazioni (progettazione e costruzione di manufatti in polietilene, supporti e adattatori per campagne sperimentali) (0.25 MU prog. + 0.25 MU off.)



2026 ongoing experiments

- MSPC, TANGO
- ALAN, ASTERIX, BOND, CROWN, HARMONIQ, SL_BETATEST, SUPERMAD
- SPHINX, QUARTET



GRANT GIOVANI 2025 - SIGLA IN CORSO in CSN5 (INFN)

Istituto Nazionale di Fisica Nucleare

MSPC

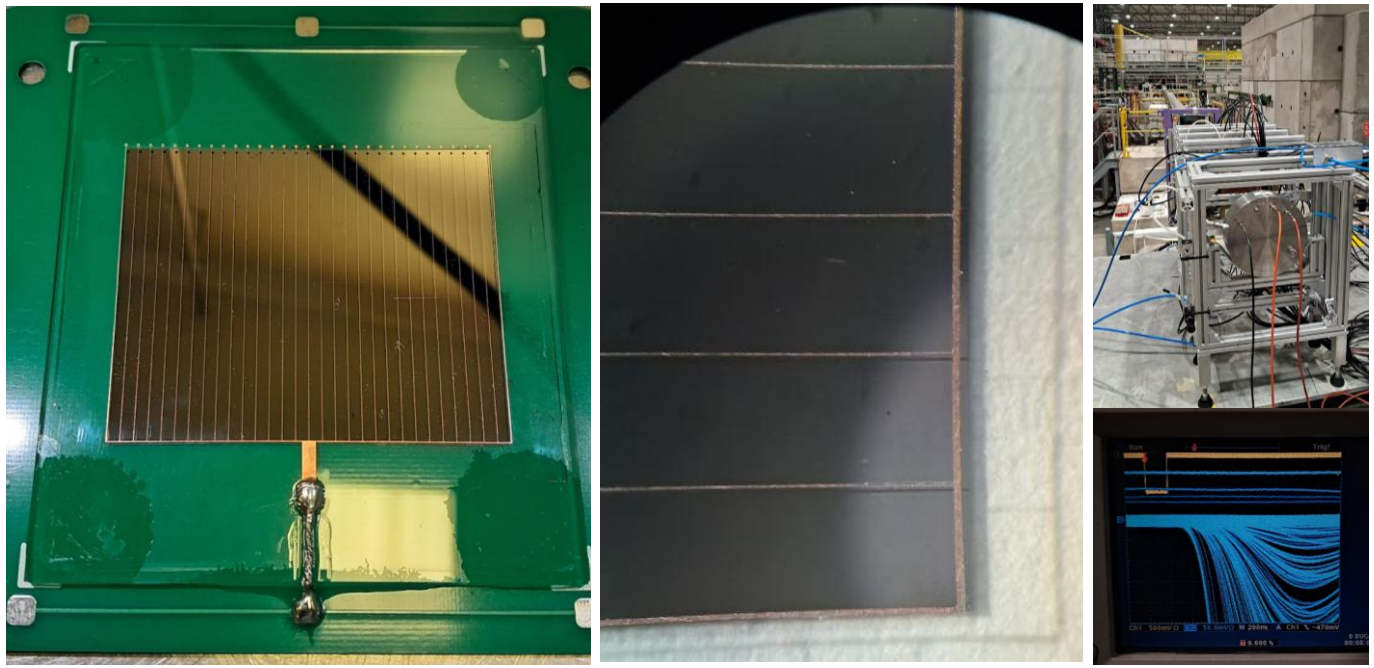
(MultiStrip Proportional Chamber)

2 anni (2026-2027)
Units: LNF
Research line: Detector Development

Matteo Giovannetti
(Resp. Naz. e Loc.) – LNF

FTE 3.15@LNF

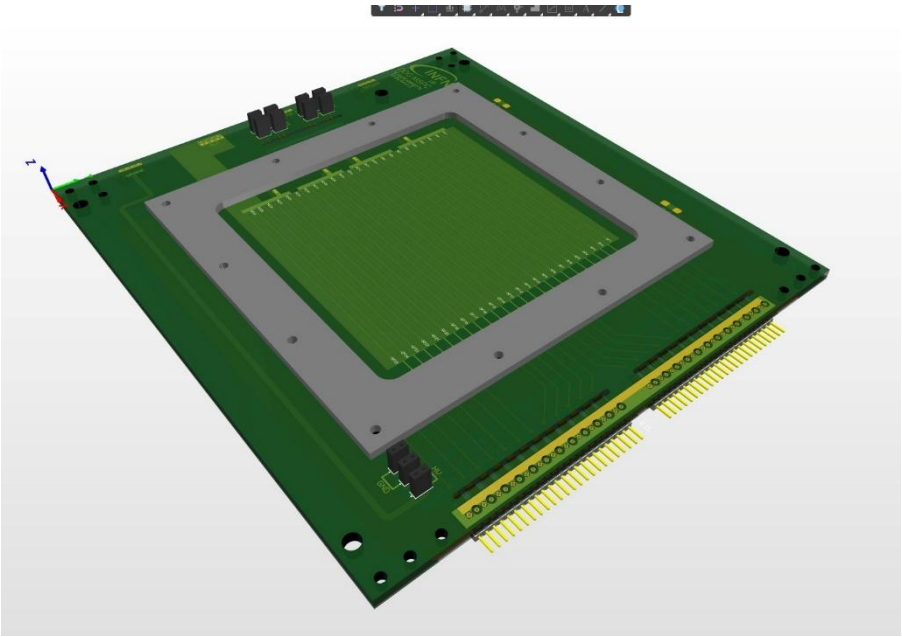
FTE tot 3.15



The main goal of this experiment is to develop a **new gaseous particle detector** for all applications where Multi-Wire Proportional Chamber would be still valid but not feasible due to loss in expertise or high costs (person-power). The detector will have **microstrip (30um) with a pitch of 2mm, build with photolitographic tecquiquest mutuated from MPGD manufacturing.**

2026 Milestones: M1, M2, M3

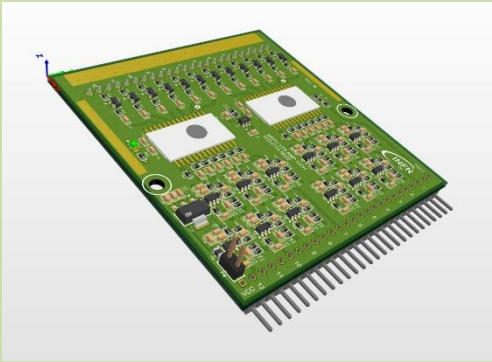
Milestone M1: amplification stage design for small area detector [100%]



Milestone M2: Production of 16 small area amplification stage [50%]

We are discussing and testing the manufacturing procedure (e.g. DLC resistivity, Kapton thickness, 30um strip width tolerance).
4 detectors will be build (active area 10x10cm2)

Milestone M3: Development of the VTX FEE board and R/O chain [50%]



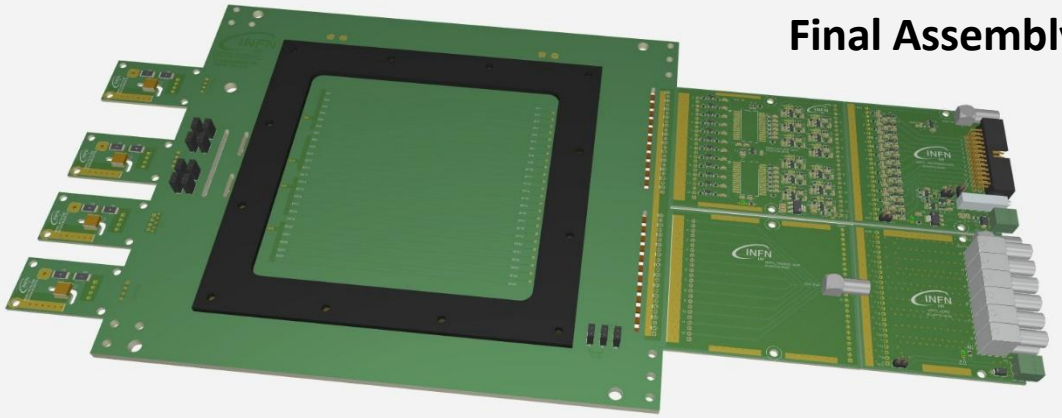
FEE design completed, waiting for component delivery.

- Each board is a 12ch. amplifier (10mV/fC, rise time 5ns)
- Additional board for processing: active and passive SUM, discriminator, LVDS translation

Design of the detector completed in the SEA-LNF service

- Active Area: 10x10cm2
- Base material: DLC+Kapton+Cu
- Amplification: 30um Cu strips, pitch 2mm. The active area is segmented in 4 sectors.
- # of channels: 24 (2 strips read together)
- # of electrodes: Cathode, Strip, DLC, (optional) backplane for bias

Final Assembly



Plan and Financial Request@LNF for 2027

Remaining design phases **[by mid of 2027]:**
- simulation of the detector signal (COMSOL, Garfield++)
- design of 20x20cm2 detectors (from the 10x10 experience)

Milestones for 2027

M4: Software for characterization of small area detectors **[April 2027]**

D1: Production of large area detectors **[July 2027]**

M5: Beam test result presented in conference/workshop **[December 2027]**

	Descrizione	Richieste (k€) (+IVA)	Totale Richieste (k€) (+IVA)
consumo	Development of large area vessel	2.2	34.0
	HV distribution for large area	1.1	
	Base material and production of large area detectors	30.7	
missioni	Mission for Beam Test crew	12.0	17.1
	Mission for Beam Test transport	2.6	
	Mission for conference	2.5	
inventario	HV board caen	7.7	7.7
Totale			**58.8**

LNF	FTE
Giovannetti Matteo (Resp. Naz.)	1.0
Zunica Gianluca	1.0
Bencivenni Giovanni	0.3
Poli Lener Marco	0.2
Morello Gianfranco	0.2
De Simone Patrizia	0.2
Felici Giulietto	0.15
Beretta Matteo	0.1
Tot./Unit	**3.15**

Collaborations

The only INFN section involved in the MSPC work is LNF. The development of the detector is carried on with the support of the CERN MPW workshop, the leading workshop in development of MPGDs technologies (Kapton etching, patterning of microstructures). The base material (Cu-Kapton-DLC foild) is produced with the CERN-INFN DLC magnetron sputtering machine, in the CERN MPT workshop.

Conference talks related to MSPC

- Matteo Giovannetti (Resp. Naz.): abstract submitted and accepted for MPGD2026 conference (09/2026)

Invited Talks related to MSPC

- None

Publications related to MSPC

- none

TANGO_RD

Toward A New Generation Of RPC Detectors

(2026-2028)

Research line: detector development

Resp. Nazionale: **B. Liberti (RM2)**

Resp. Locale LNF: **D. Piccolo**

FTE 1.3@LNF

FTE tot 2.8



- *ToV:* **B. Liberti (PI)**, A. Rocchi, E. Vallocchia **(1.2 FTE)**
- *LNF:* D. Piccolo, S. Bianco, A. Paoloni, S. Meola, G. Saviano, C. Venditozzi **(1.3 FTE)**
- *Bari:* M. Abbrescia, M. De Serio, A. Pastore **(0.3 FTE)**
- Totale **2.8 FTE**

The TANGO_RD project aims to bring the RCC technology to a maturity level to be used for future experiments. The main objectives are:

1) to systematically characterize the working principles of the RCC detector as a function of geometrical construction parameters,
2) to study the performance with different ecological gas mixtures and at high over-pressure
3) to explore innovative materials to enhance the rate capability
4) to study the scalability of the technology for large size applications.

2026 Milestones: percentage under discussion with referees

M1.1: Material procurement for a complete set of tubes with different geometries (thicknesses, gas gap, radii)

Advanced Material available for 2 RCCs with different geometries

Order in progress for the material of additional 6 RCC geometries

M1.2: Production of first RCC glass prototypes

Advanced First prototypes almost finalized

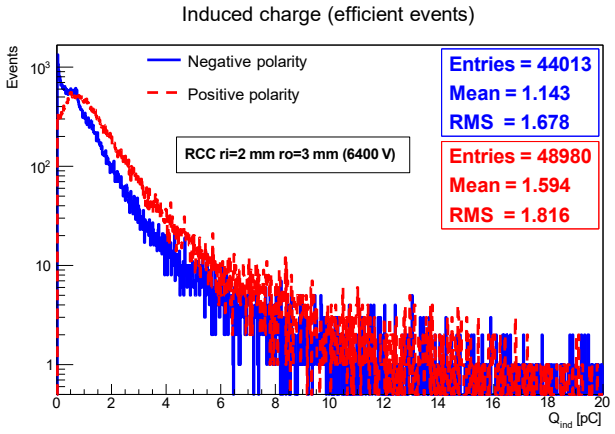
M2.1: Comparison data/MC of charge distributions for std gas mix vs geometrical parameters

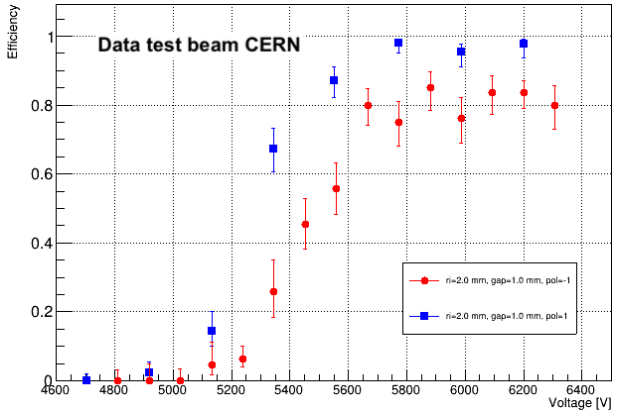
Advanced Simulation framework available. First comparisons performed with old data

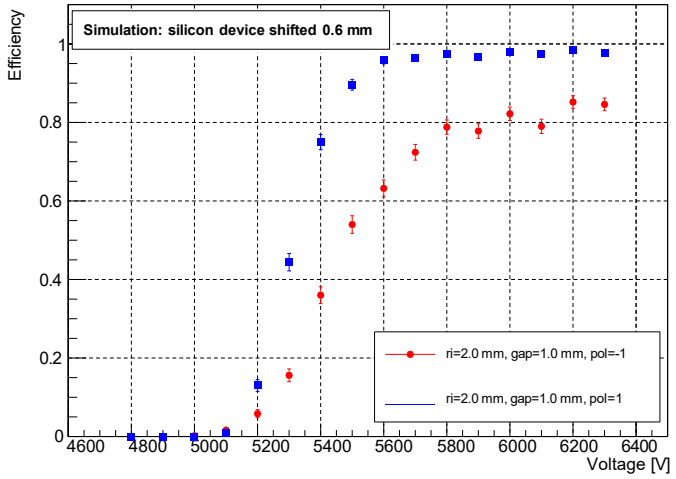












Plan and Financial Request@LNF for 2027: PRELIMINARY

	2027					
		ROMA 2	LNF	Bari	Tot (KEuro)	
Consumi	Materiale per prototipi LP e HP				10	5
	Gas per test beams su singoli tubi				3	
Totali consumi 2027					13	
Inventariabile	Alimentatore HV +30 kV				1,5	1,5
Totali inventariabile 2027					1,5	
Missioni	Test singolo tubo e prototipi con nuovi materiali a BTF				3	
	Test singolo tubo e prototipi con nuovi materiali a GIF++				10	
Totali missioni 2027					13	
Totale 2027					**27,5**	

LNF FTE 2027

LNF	FTE
D. Piccolo (Resp loc)	0.3
S. Bianco	0.2
A. Paoloni	0.1
S. Meola	0.3
C. Venditozzi	0.2
G. Saviano	0.2
Tot./Unit	1.3

LNF service requests 2027

1.5 MU Detector Development and Experimental Activities Support

2027 Milestones: PRELIMINARY

M1.3 June 2027 Validation of the quenching principle and Characterization of performance vs geometric parameters with std mix for RCC

M4.2 June 2027 Flexible chamber construction for housing different electrodes

M5.1 June 2027 Layout and mechanical design of two prototypes: LP type and HP type

M2.2 December 2027 geometrical simulation of LP and HP prototypes

M3.2 December 2027 Characterization of performance as a function of overpressure

Publication

A standalone simulation program for resistive cylindrical chamber (RCC)

Eur. Phys. J. C (2026) 86:351.

https://doi.org/10.1140/epjc/s10052-026-15576-5

Eur. Phys. J. C (2026) 86:351
https://doi.org/10.1140/epjc/s10052-026-15576-5

THE EUROPEAN PHYSICAL JOURNAL C

Regular Article - Experimental Physics

A standalone simulation program for resistive cylindrical chamber (RCC)

M. Abbrescia[1,2], S. Colafranceschi[3,4], M. De Serio[1,2], B. Liberti[5], S. Meola[3,6], A. Paoloni[3], A. Pastore[1], D. Piccolo[3,a], A. Rocchi[3,5], G. Saviano[3,7], C. Venditozzi[3,7]

[1] INFN Sezione di Bari, Via E. Orabona 4, 70125 Bari, Italy
[2] Dipartimento Interateneo di Fisica, Università degli studi di Bari, Via Amendola 173, 70125 Bari, Italy
[3] INFN, Laboratori Nazionali di Frascati, 00044 Frascati, Italy
[4] James Madison University (JMU), Harrisonburg, VA 22807, USA
[5] INFN, Sezione di Roma Tor Vergata, Via della Ricerca Scientifica 1, 00133 Rome, Italy
[6] Università degli Studi Guglielmo Marconi, 00193 Rome, Italy
[7] Dipartimento di Ingegneria Chimica Materiali Ambiente, Università di Roma La Sapienza, Via Eudossiana 18, 00184 Rome, Italy

Received: 24 October 2025 / Accepted: 18 March 2026
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Abstract In recent years, the resistive cylindrical chamber (RCC) has been introduced as a novel gaseous detector, extending the well-established resistive plate cham- A first comparison with available RCC experimental data is also presented, providing an initial assessment of the model performance in realistic operating conditions.

Esperimento in CSN5 (INFN)

Istituto Nazionale di Fisica Nucleare



ALAN
(Active pLAsma for beNding)

3 anni (2026-2028)
Units: LNF, INFN-Milan
Research line: Accelerators

Angelo Biagioni (Resp. Nazionale)

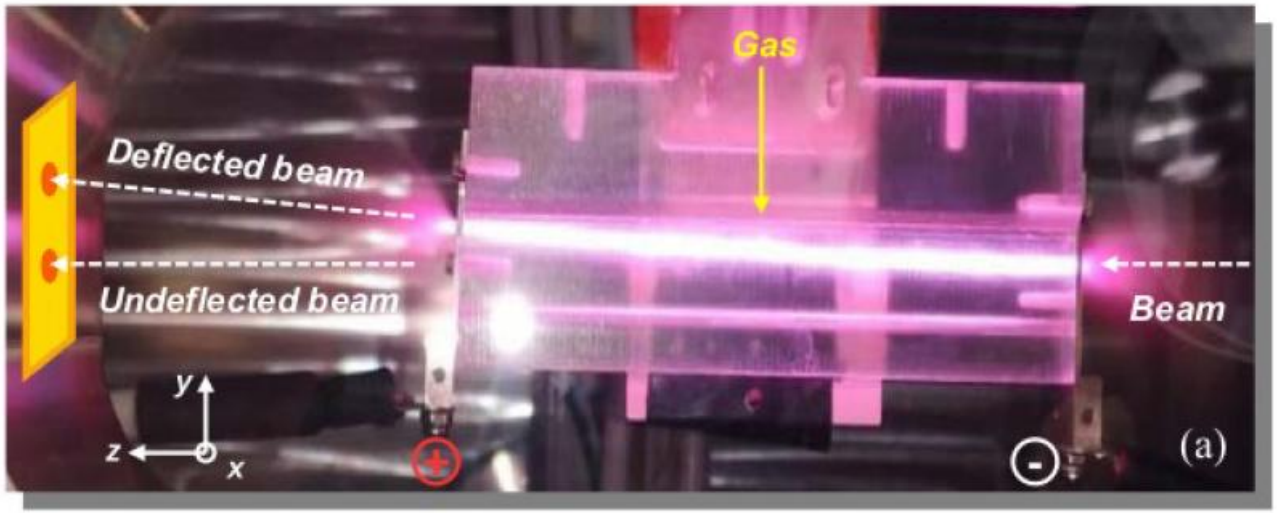
Angelo Biagioni (Resp. Locale@LNF)

Andrea Renato Rossi (Resp. Locale@INFN-Milano)

FTE 1.9@LNF

FTE 1.05@INFN-Milano

FTE tot 2.95 (2026)



The ALAN project is built around three main objectives:

- Developing an innovative plasma-based bending device.
- Developing suitable materials for the gas-filled discharge capillary to confine plasma generated by high-voltage discharges.
- Developing high-voltage sources capable of delivering the discharge currents required for particle beam bending.

Milestone_2026 ALAN project

M.1: Design of the curved capillary
M.2: Plastic prototype
M.3: Studies on beam-plasma interaction

M.3 Studies on beam-plasma interaction

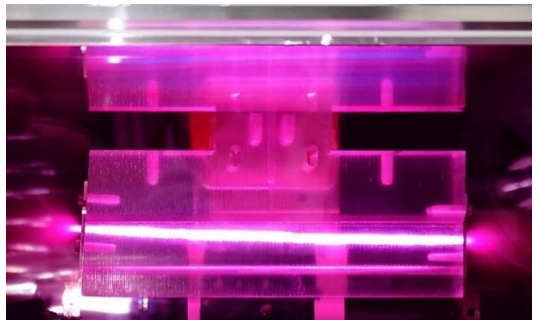
$$B_\phi(r) = \frac{\mu_0}{r}\int_0^r J(r')r'dr'$$

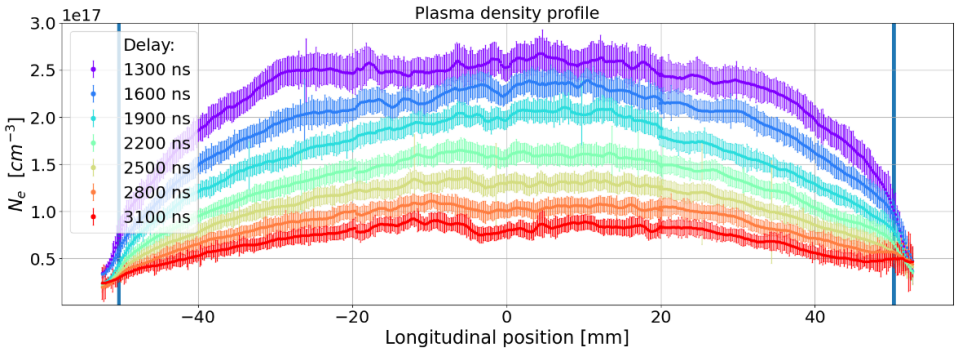
B ~ 1,4 T
I_0 ~ 7 kA
R_{cap} = 1 mm

Rc ~ 15 cm
L = 23 m
α = 90°
E = 60 MeV

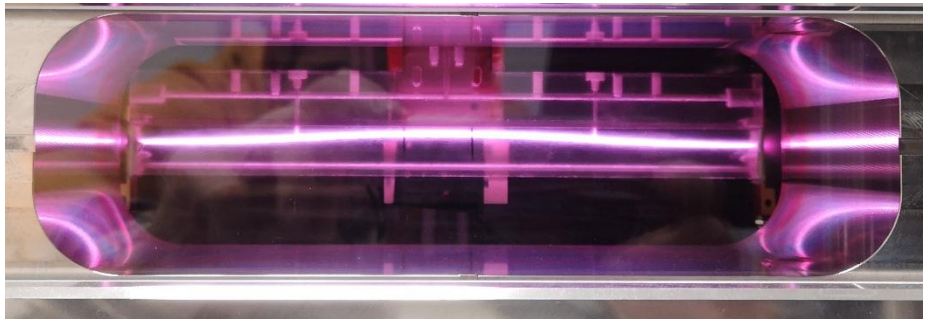
M.1 Design of the curved capillary/**M.2** Plastic prototype

1. Dipole-like solution





2. Chicane-like solution



First prototype



Rc ~ 50 cm
L = 10 cm
α = 12°
E = 60 MeV

B ~ 0,4 T
I_0 ~ 2 kA
R_{cap} = 1 mm

Project Workflow

- Fabrication of curved capillaries with single and double bending (preserving a straight beam trajectory at the screen)
- Initial low-current prototypes (2–3 kA), providing equivalent bending radii of approximately 0,3–0,5 m and generating magnetic fields of about 0.4-0.6 T.
- Study of Marx-generator circuits for generating 4–8 kA currents and magnetic fields of about 0,8–1,6 T for 12-25 cm bending radii.

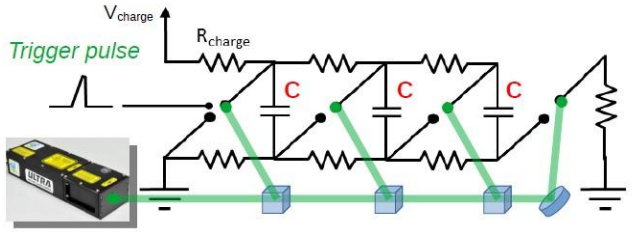
Plan and Financial Request@LNF for 2027

Milestones for 2027

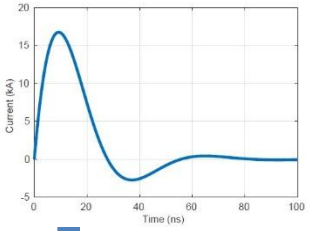
M.4: Marx generator circuits design

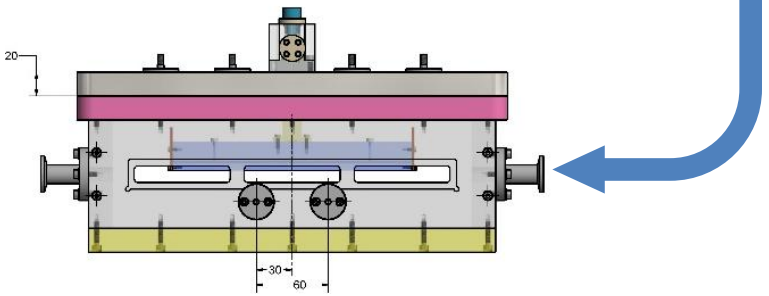
M.5: Shapal capillary design

Marx generator circuits



Current pulse





Capillary chamber to test Marx generator circuits

LNF	Descrizione	Richieste (k€)	Totale Richieste (k€)
consumo	Realizzazione Capillari in Shapal	30	30
missioni	missioni	5	5
inventario	Supporto per capillare, sistema distribuzione gas e Movimentazione capillare	30	60
	Realizzazione prototipi di Marx Generator (MG)	30	
Totale			**95**

LNF	FTE
(Resp. Naz.) Angelo Biagioni	0.3
Maria Pia Anania	0.3
Mario Galletti	0.1
Riccardo Pompili	0.1
Gemma Costa	0.1
Alessio Del Dotto	0.1
Tot./Unit	1.0

INFN-Milano	FTE
(Resp. Loc.) Andrea R. Rossi	0.35
I. Drebot	0.05
B. Paroli	0.1
N. Piovella	0.5
Tot./Unit	1.0

Nuova Proposta di Esperimento in CSN5 (INFN)

Istituto Nazionale di Fisica Nucleare

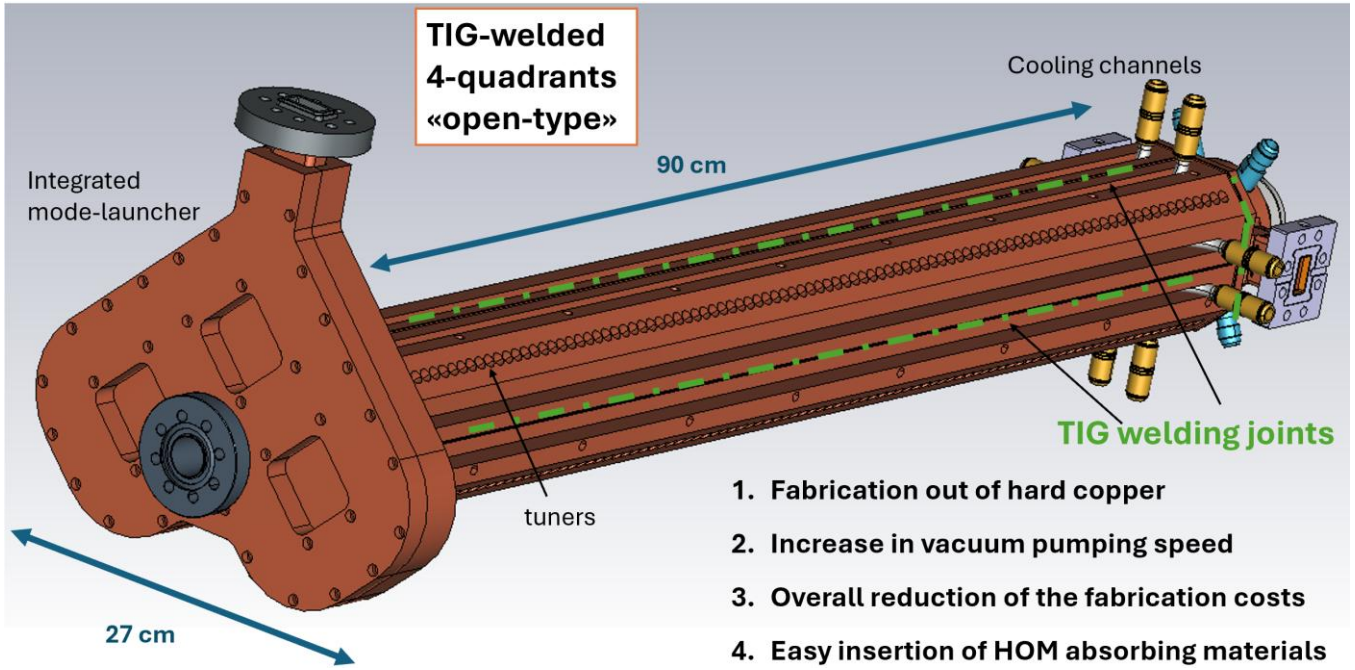
ASTERIX

(**A**ccelerating **ST**ructures made of multipl**E** secto**R**s **I**n ***X***-Band)

3 anni (2025-2027)
Units: LNF, LNS, Roma1
Research line: Accelerators and related technologies

Luigi Faillace (Resp. Naz.) – LNF

FTE 2.1@LNF

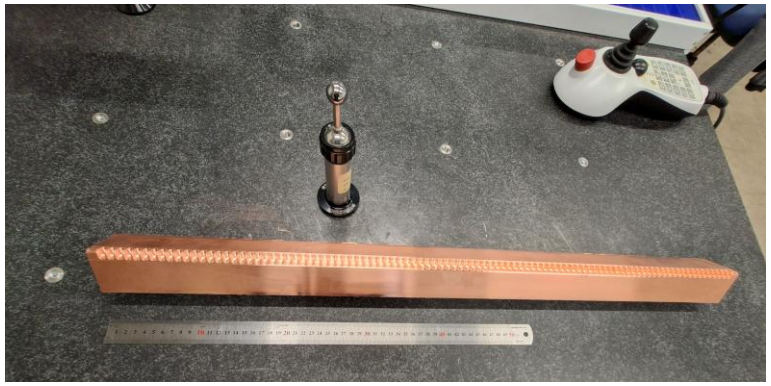


The main goal of this experiment is the design, fabrication and high RF power testing of **four-quadrants ("open-type") X-band RF accelerating structures** made of hard copper, joined and vacuum sealed by using **TIG welding** (**"braze-free" technique)** in order to achieve higher **accelerating gradients (>100 MV/m)**, higher efficiency, as well as **cost-effective and robust manufacturing**.

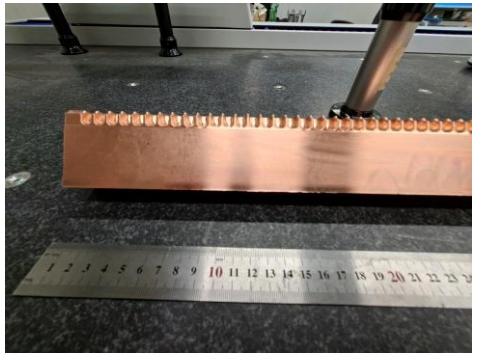
MECHANICAL Prototype 2025

During the first year of the project (2025), we have developed the mechanical drawings and engineering of the full-scale structure, without RF power couplers, in order to perform the following tests:

- TIG welding process,
- structure straightness and quadrant alignment,
- vacuum performance.

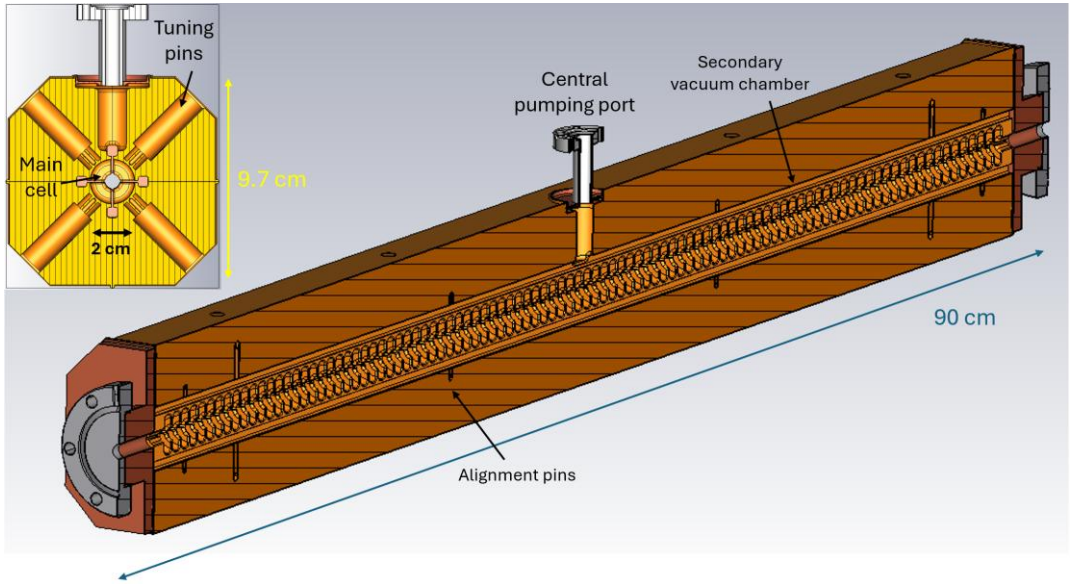








Final CAD model of the prototype including tuning pins, alignment pins, welding teeth, central pumping port and secondary vacuum chambers.



The machining process of the mechanical prototype is completed. The four quadrants were fabricated inside a CNC milling machine which can host the whole parts:

- Machine Model: Doosan DNM 750L
- Precision for the prototype <0.05mm
- Roughness for the prototype <1 Ra

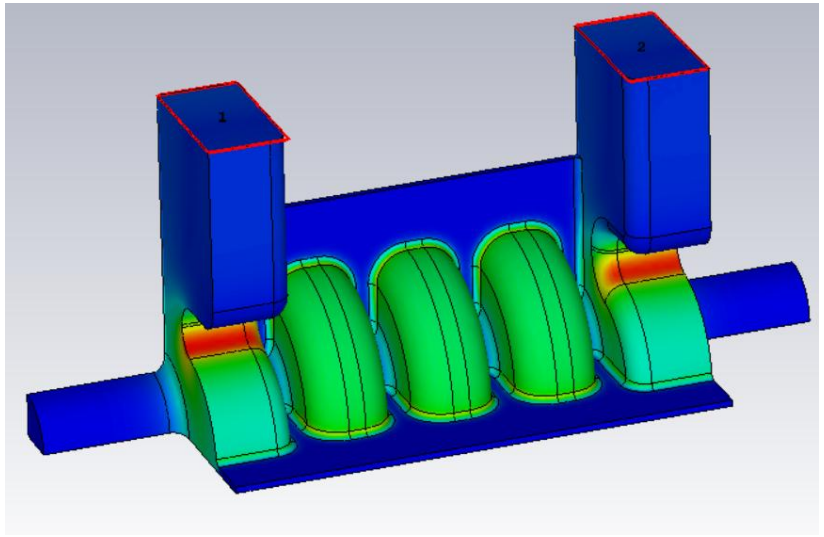
The sectors were TIG welded and inert gas was flushed during the process.

Mechanical and vacuum tests are in-progress!

2026 Milestones: M3 and M4.

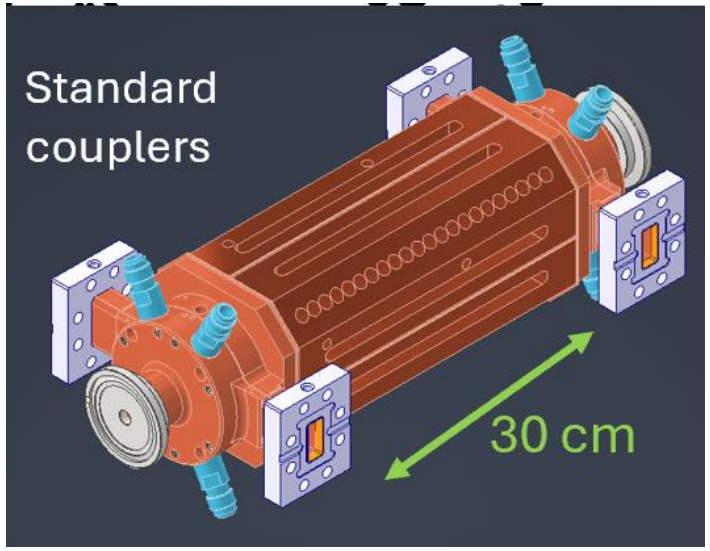
Milestone M3: Fabrication of small-scale prototypes/Mechanical validation [September 25, 2026]

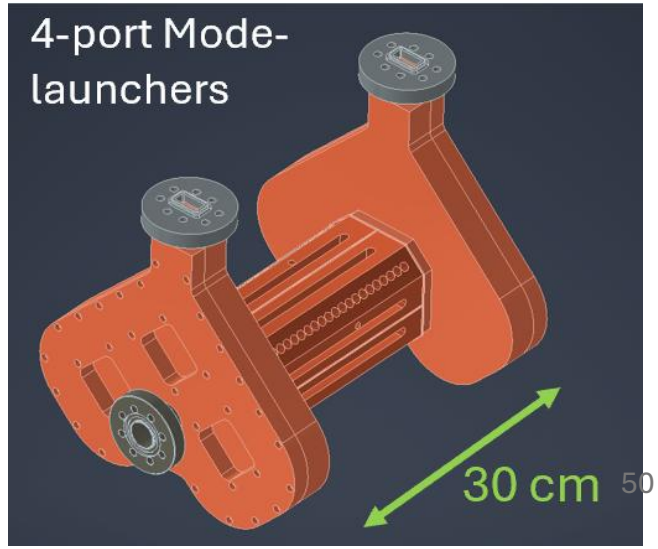
- The procurement has started;
- Order for two structures soon to be issued.



Parameter	Value
Frequency [GHz]	11.994
Gradient [MV/m]	100
Input power [MW]	77
Operating mode [rad]	$2\pi/3$
Cell length [mm]	8.333
Cell Number	120
Full structure length [m]	1
Disk thickness [mm]	2
Quality factor	6931
Iris radius [mm]	3.5
Shunt impedance [MΩ/m]	97.3
Group velocity [c%]	0.027
Filling time [ns]	123
Attenuation factor	0.6545
Peak S_c [W/μm^2]	8.8
Pulse Heating [K]	24.6
E_p [MV/m]	245

Milestone M4: Low and High RF power testing of small-scale prototypes [December 31, 2026]





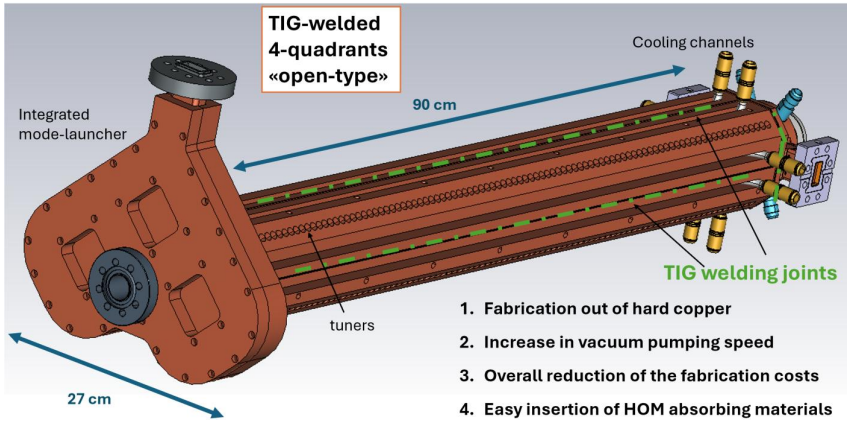
Plan and Financial Request@LNF for 2027

- Characterization of the main RF parameters of the full structure will be carried out by using a Vector Netwok Analyzer at the LATINO Lab.
- The RF high power tests will be performed at TEX to quantify the structure high-gradient performance in terms of the maximum achievable accelerating gradient, the RF breakdown probability and the dark current emission..

Milestones for 2027

M5: Fabrication of final full-scale structure and Mechanical Validation [July 2027]

M6: Low and High RF power testing of final full-scale structure [December 2027].



	Descrizione	Richieste (k€)
consumo	Approvvigionamento di struttura finale in rame, lunghezza 90 cm complete di tutti i supporti: 1 cavita' da 120 celle, piu' coupler di potenza RF.	150
missioni	Missioni per misure (comprese la caratterizzazione meccanica) presso l'azienda produttrice, per la supervisione e il monitoraggio della fabbricazione della struttura. Missioni per due persone presso LNS e Roma1 per supervisionare caratterizzazione RF a bassa potenza di accoppiatori (RF couplers) realizzati precendentemente dai colleghi e propedeutici alla progettazione degli RF couplers di ASTERIX.	4
Totale		**154**

LNF	FTE
L. Faillace (Resp. Naz.)	0.4
D. Alesini	0.1
M. Bellaveglia	0.15
S. Bini	0.2
F. Cardelli	0.1
E. Chiadroni	0.2
A. Falone	0.1
A. Gallo	0.15
A. Giribono	0.1
A. Liedl	0.1
L. Piersanti	0.1
S. Pioli	0.2
B. Spataro	0
G. Silvi	0.1
C. Vaccarezza	0.1
Tot./Unit	**2.1**

Collaborations

We are currently hosting, since October 2024 at INFN-LNF, a PhD student form University of Science and Technology of China, Zhicheng Huang, who is helping on the RF design and Wakes/HOMs optimization of the ASTERIX structures for both single- and multi-bunch operation modalities.

Contributed Talks in 2026

- Luigi Faillace (Resp. Naz.): "ASTERIX: Development of TIG-Welded Four-Quadrant X-Band RF Accelerating Structures at INFN." Advanced Accelerator Concepts AAC2026 at UCLA, Los Angeles.

Publications/Abstracts in 2026

Article accepted for publication on Peer-review Journal:
- L. Faillace et al., "Compact, cost-effective, advanced electron accelerators", Journal of Advanced Instrumentation in Science (JAIS) after the Invited Talk I gave at the African Conference on Fundamental Physics and Applications (ACP 2025).

Two proceedings at the International Particle Accelerator Conferece IPAC26 in Deauville, France:
- L. Faillace et al., "Characterization of the four-quadrant X-band cavity prototype for the ASTERIX project".
- M. Mostajeran, "Study Of Multipactor Discharge In The X-band Resonator For The Asterix Project".

Abstract
- T. Abe (KEK) et al., High-Power Test Results for Multiple-Sector X-band Accelerating Structures Fabricated Using Different Materials but Based on the Same Design, 13th International Workshop on the Mechanisms of Vacuum Arcs (MeVArc 2026)

CROWN - CSN5 (INFN)

Istituto Nazionale di Fisica Nucleare

CROWN

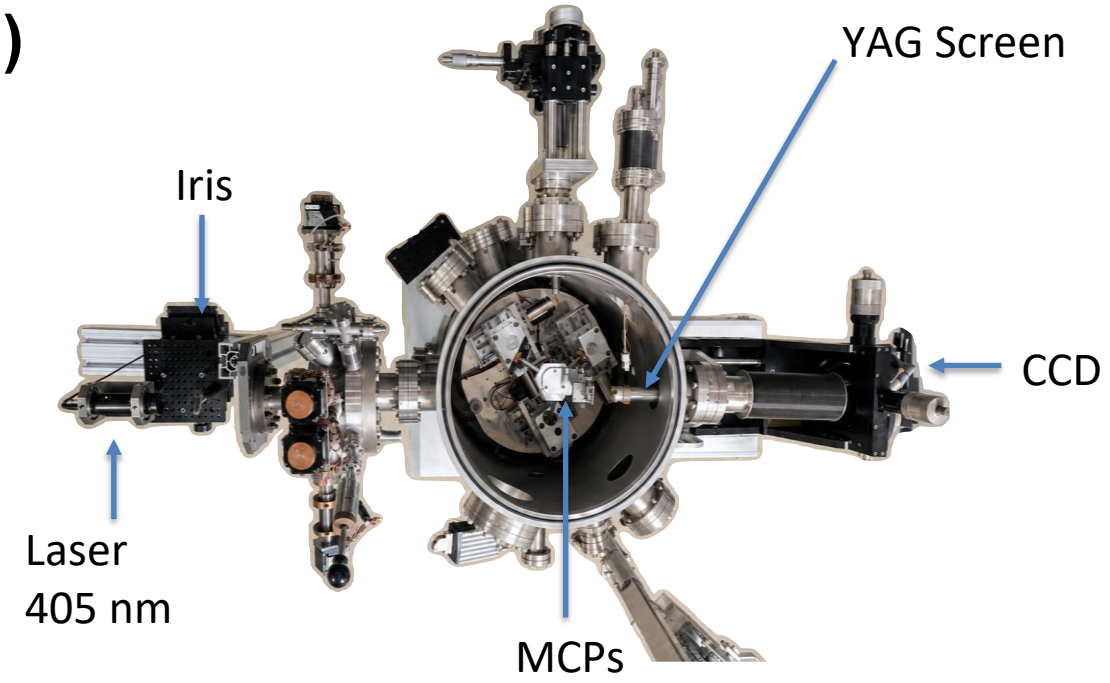
(Confocal microscopy at sOft X-ray WaveleNgths)

4 anni (2023-2026)
Research line: Accelerators and related technologies

Monica de Simone (Resp. Naz.) – Trieste

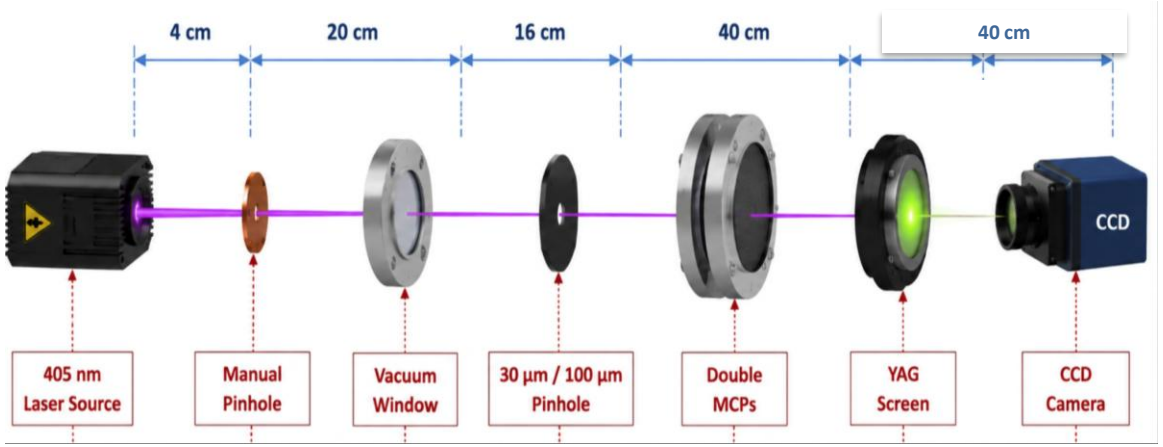
FTE 2@LNF

FTE tot 3.2



The main components of the optical set up

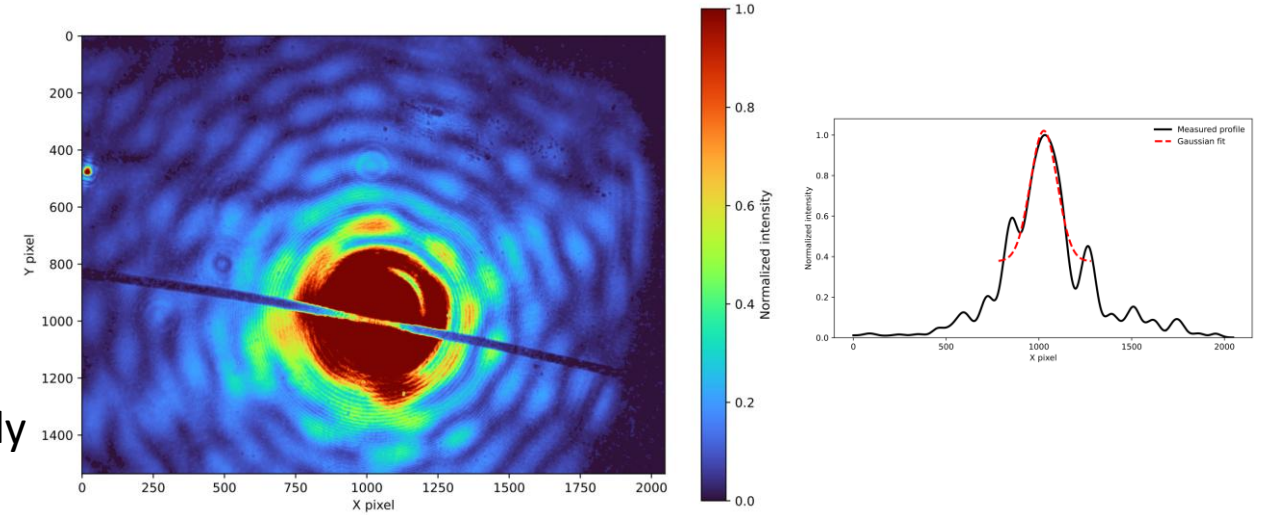
In this proposal a VUV/X-ray microscope layout based on the development of a confocal microscope using bent meta-lenses, i.e., Micro-Channel Plates (MCPs) will be pursued. The main goal of this experiment is to characterize the response of double MCPs to 405 nm laser light using 30 µm and 100 µm pinholes. In parallel, the software control of the Hexapod and Hexant manipulators was tested, in order to verify reliable motion control, alignment capability, and reproducibility of the experimental setup. In the last six months of the project the microscope will be tested with different wavelength.



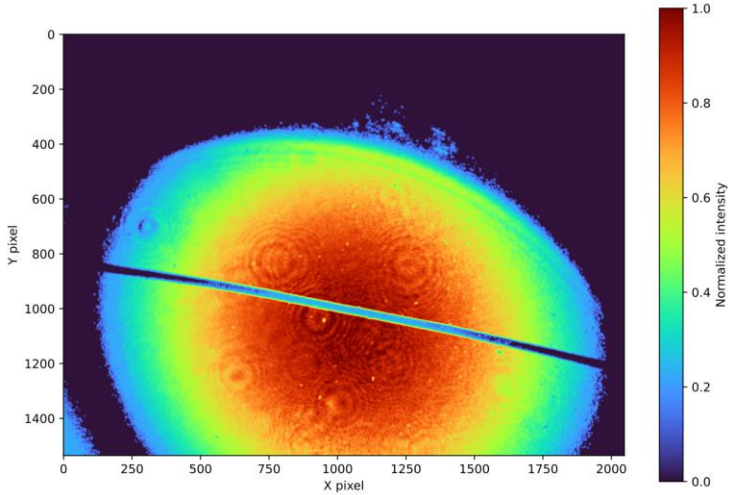
The optical layout for the tests

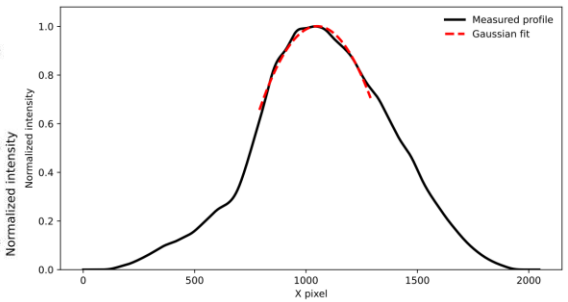
The 405 nm laser beam was recorded without MCPs, directly and after 30 µm/100 µm pinholes. Horizontal profiles were extracted and Gaussian-fitted for approximate beam-width comparison.

Direct 405 nm laser spot

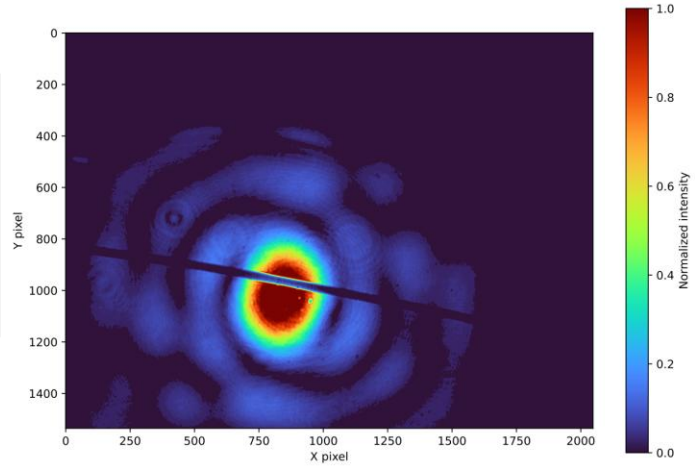


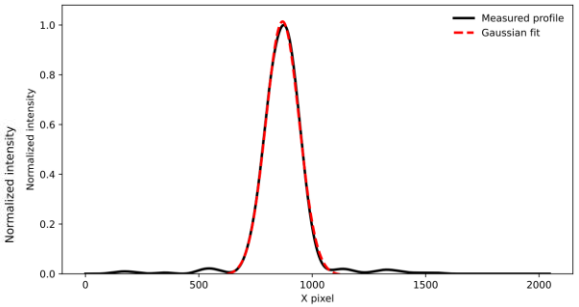
30 µm pinhole reference beam





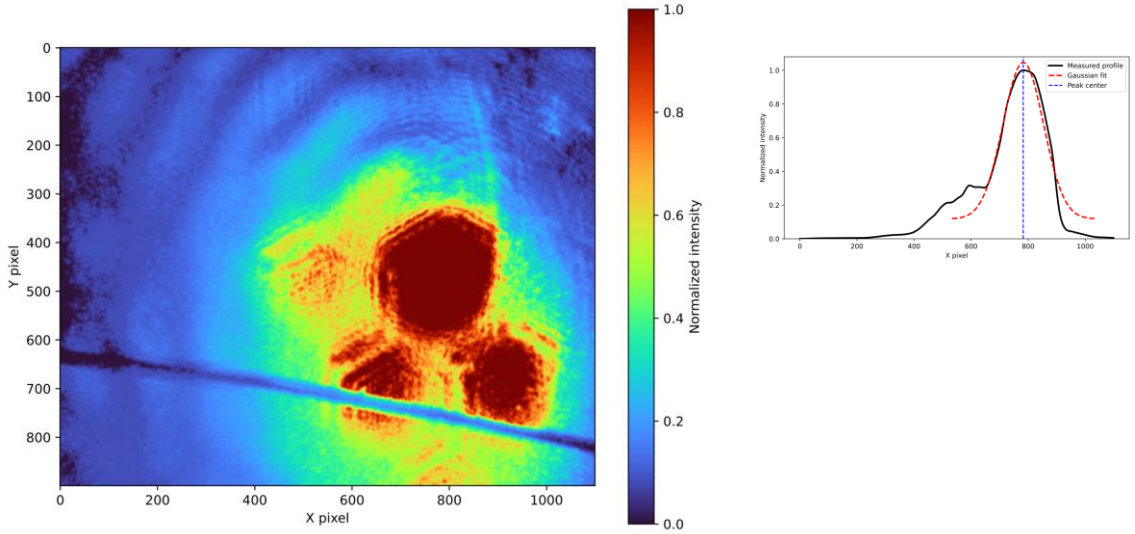
100 µm pinhole reference beam



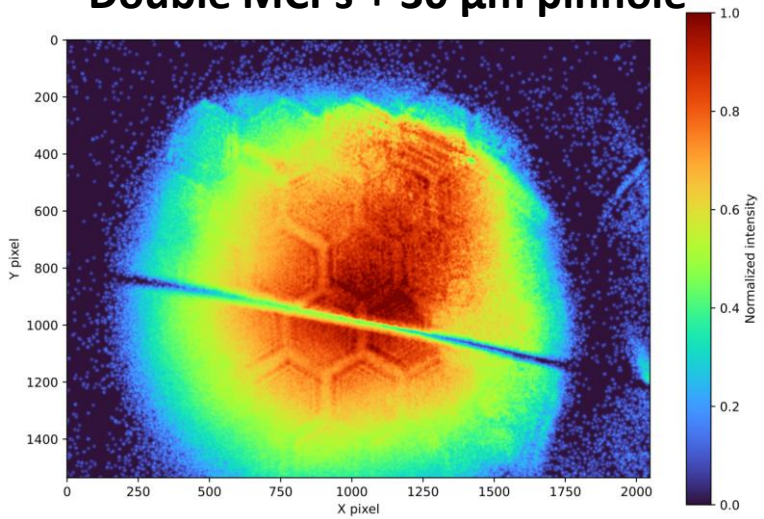


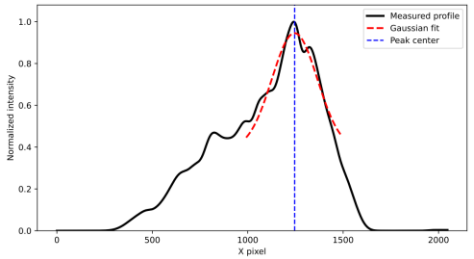
laser @405 nm

Double MCPs – no pinhole



Double MCPs + 30 µm pinhole

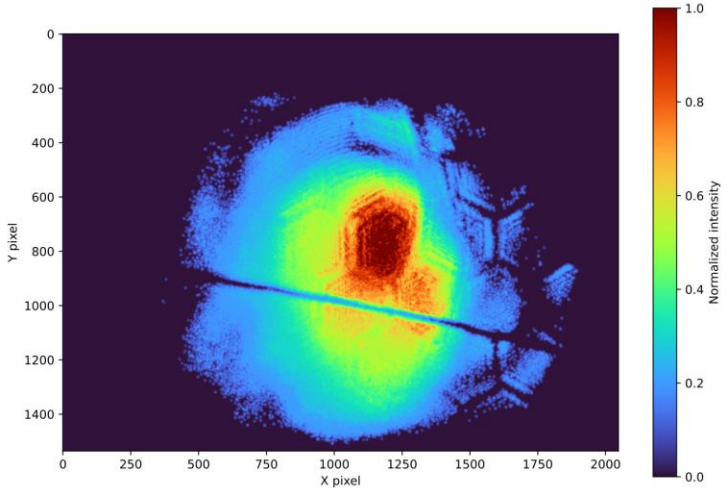


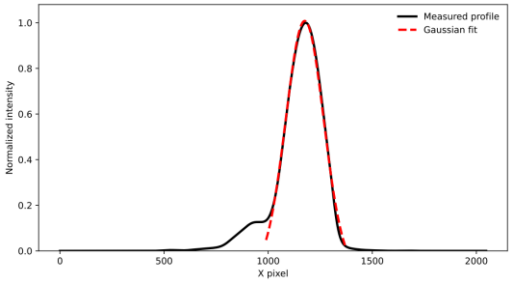


The filtered images reveal a clear hexagonal-like transmission pattern produced by the double MCP channel structure.

Horizontal profiles were extracted to compare the beam envelope for no pinhole, 30 µm, and 100 µm pinhole conditions.

Double MCPs + 100 µm pinhole





Activity foreseen for the last 6 months

In the last 6 months of the experiment we will strive to fulfill the final milestone of the project, testing MCP at shorter wavelengths: *Characterization of the microscope and acquisition of images with the 2D detector. Spatial resolution tests using a laser at 266 nm (~4 eV) and with synchrotron radiation (SR) in the soft x-ray range (if available).*
Unfortunately the delay in the start of Elettra 2.0 makes it impossible to exploit SR for this goal. We are thus working to exploit a Nd:YAG laser (Edgewave, model Innoslab II) to illuminate the MCP optics with radiation at 266 nm, and with VUV light below 100 nm, produced via a prototype gas puff source [P.W. Wachulak 2016, Rev. Sci. Instrum. 87, 91501, doi: 10.1063/1.4962012].

Collaborations

Dr. Rebeca-Martinez-Vaquez group at CNR-Istituto di Fotonica e nanotecnologie (Milano- Italy)

Publications

M.I. Mazuritskiy, A.M. Lerer, A. Marcelli, Z. Ebrahimpour, S. B. Dabagov,
Diffractive patterns generated by a couple of microchannel plates to characterize synchrotron radiation sources,
Nuclear Instruments and Methods in Physics Research Section A: Accelerators, Spectrometers, Detectors and Associated Equipment,
Volume 1075, 2025, 170415;

Z. Ebrahimpour, M. Coreno, S. Dabagov, M. de Simone, M. Mazuritskiy, A.E.A. Mohamed, J. Rezvani, F. Zuccaro, A. Marcelli
Advancing soft X-ray microscopy with diffractive metalenses: Experimental design and initial results,
Nuclear Instruments and Methods in Physics Research Section A: Accelerators, Spectrometers, Detectors and Associated Equipment,
Volume 1083, 2026, 171181;

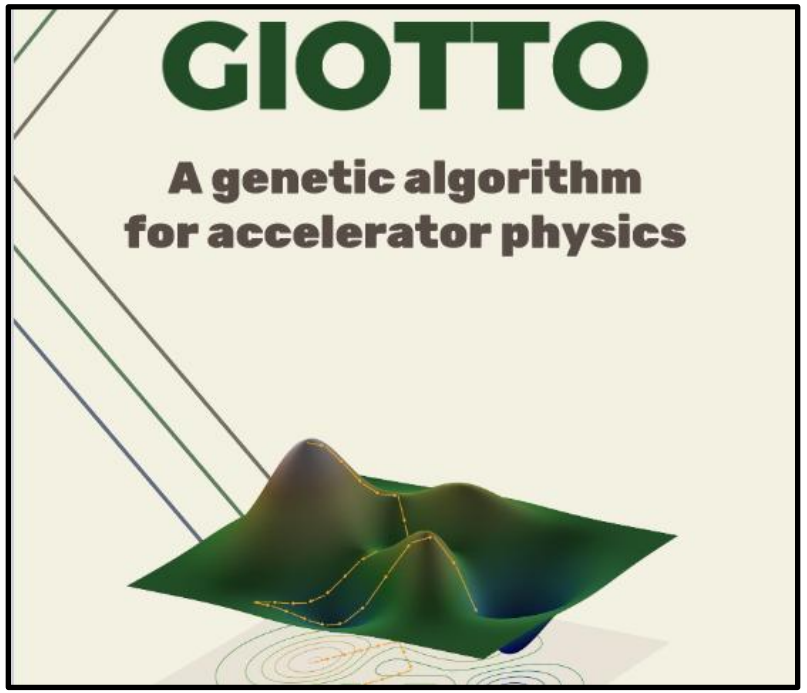
A. Muqeet, A. Gabriela Ciriolo, E. Principi, M. Coreno, M. de Simone, S. Orlando, C. Vozzi, J. Turnšek, K. Bučar, A. Mihelič,
R. Osellame, M. Žitnik, R. Martínez-Vázquez
Laser fabricated microfluidic glass chip to study amplified spontaneous XUV emission of helium gas
Proc. SPIE PC13880, Laser Applications in Microelectronic and Optoelectronic Manufacturing (LAMOM) XXXI,
PC138800D (5 Mar 2026); https://doi.org/10.1117/12.3080695

Esperimento in CSN5 (INFN)

GIOTTO

(Genetic Interface for OpTimizing Tracking with Optics)

3 anni (2026-2028)
Units: LNF, LNS, Sez. MI
Research line: Accelerators



Anna Giribono
(Resp. Loc.) – LNF

FTE 1.4@LNF

FTE tot 5.3

The main goals of this experiment is:

- Upgrade and establishment of GIOTTO as an institutional optimization tool, developed within INFN over the years
- Enable **chained optimizations** with: HFSS/ANSYS, CST, BuildCAvity/SuperFish; main outcomes for EuPRAXIA optimization, FCC-ee positron capture, Dielectric accelerations.
- Integrate **GIOTTO into accelerator Control Systems:** now nearly ready for STAR, open discussion coming soon @LNF

2026-2027 Milestones M1-M2 done @ Milano, M8 next year @ LNF

Month	Task	3	6	9	12	15	18	21	24	27	30	33	36	
WP0	T 0.1													
WP1	T 1.1			♦M2										GIOTTO_CL Classical Dev.
	T 1.2		♦M1											GIOTTO_CS Contro System Dev.
	T 1.3						♦M7							Jitters sensitivity map study & development
	T 1.4					♦M5			D1					RF Cavity design and BD seamless optimization (BuildCavity)
WP2	T 2.1						♦M8					D4		GIOTTO_CS test on @ STAR,SPARC,HARMONIQ
	T 2.2									D2				Jitters analysis and optimization on WPs @ SPARC
WP3	T 3.1					♦M6				D3				Cavity design and BD seamless optimization (HFSS/ANSYS)
WP4	T 4.1													User-friendly interface
	T 4.2				♦M3								D5	Dedicated website
	T 4.3										♦M9			Training cours
	T 4.4				♦M4								D6	Giotto Manua
♦ proj.		1y	♦	♦	♦♦	2y ♦♦	♦	♦		3y	♦			

Milano

M1 (for INFN-MI) – GIOTTO_Classic	α-version in use @ LNF
M2 (for INFN-MI) – GIOTTO_EPICS	α-version in use @ STAR CS

Defended in April 2026



LNF

At the end of 2025, CSN5 advanced €30k to buy two workstations for GIOTTO @ LNF, which have recently been installed.
M8 (for LNF in 2027) – GIOTTO Golden Orbit tests SPARC and/or STAR

Plan and Financial Request@LNF for 2027

- Activities on GIOTTO_EPICS for advanced Control System tests, such as automatic Golden Orbit setup (GIOTTO vs BPM and optics control), STAR/SPARC

- Activities on the GIOTTO Classic version for EuPRAXIA, optimizing beam dynamics (BD) together with cavity design (HFSS/ANSYS).

- If ready, first tests on the HARMONIQ focusing channel (many parameters to be optimized using GIOTTO_EPICS).

Milestones for 2027, M8 as cited in the previus slide

	Descrizione	Richieste (k€)	Totale Richieste (k€)
consumo	0	0	0
missioni	3–4 technical meetings in Milan (1–2 people) and/or at STAR (UNICAL) to work on GIOTTO_EPICS at SPARC and/or STAR. Furthermore, possible setup of optimization runs for the SPARC Golden Orbit (EPICS on SPARC) and/or complex optimization for EuPRAXIA (HFSS with BD). 1 shift @ STAR on GIOTTO_EPICS	5	xxx
inventario	xxx	xxx	Xxx
	xxx	xxx	
	xxx	xxx	
Totale			**5**

LNF	FTE
A. Giribono (Resp. Loc.)	0.2
L. Faillace	0.1
L. Pellegrino	0.1
L. Piersanti	0.1
G. J. Silvi	0.3
A. Stella	0.1
C. Vaccarezza	0.1
A. Vannozzi	0.1
A. Del Dotto	0.1
N. Lo Gullo	0.2
Tot./Unit	**1.4**

Collaborations

About GIOTTO, we are currently collaborating in many different directions. At STAR (UNICAL), it is already integrated into the Control System (CS), and many more tests have been carried out driving a LINAC simulator under the CS. We have just changed the names of the EPICS PV variables to move to the real machine. We haven't done so yet because the S-band STAR klystron has been under maintenance since April.
These tests at STAR should open, asap, the discussion to perform tests also at SPARC_Lab.
At the same time, we are ready to start optimizing a focusing channel in the framework of the HARMONIQ experiment (in synergy with GIOTTO, LFN, RN. A. Vannozzi), which requires the optimization of many parameters. It is a key component for all plasma-based acceleration techniques. There is also an interesting collaboration, established by the Milano group, with DESY (F. Lemery, K. Floettmann) on the Cherenkov FEL, which has already been optimized using GIOTTO.
Then, we are ready to start the study of beam-dynamics optimization coupled with cavity-design FEM software (in chain); using such as HFSS/ANSYS, CST, and BuildCavity (based on SuperFish).

Invited Talks related to GIOTTO

- G. J. Silvi (in the LNF GIOTTO team): Invited at UBA26 (Candle, Yerevan, Armania) for "well-known experties in the field of Beam Dynamic Optimization for Plasma Accelerators high brightens injectors“ – Activities related to the optimization studies performed by Gilles within GIOTTO

Publications (GIOTTO Milano Group)

‘Sub-harmonic bunchers for a low-energy ERL injector: the BriXSinO approach’
This is a study mainly based on GIOTTO's capabilities, accepted by NIM-A in June 2026. Since the work builds on previous GIOTTO activities, the LNF contribution was not yet in place at that time

Proposta di Esperimento in CSN5 (INFN)

Istituto Nazionale di Fisica Nucleare

HARMONIQ

Halbach **A**rray for **R**otational **MO**dulation of **N**dFeB **I**ntegrated **Q**uadrupoles

3 anni (2026-2028)

Units: LNF,MI,NA

Research line: Accelerator

Vannozzi Alessandro (Resp. Naz) – LNF

FTE 2 @LNF

FTE 2,8 tot



The main goal of HARMONIQ is to realize and test TPMQ's based on QUAPEVA design but optimized in terms of **compactness** and **reduced multipolar skew components for beam strong focusing application** starting from the Plasma Acceleration case (most challenging)

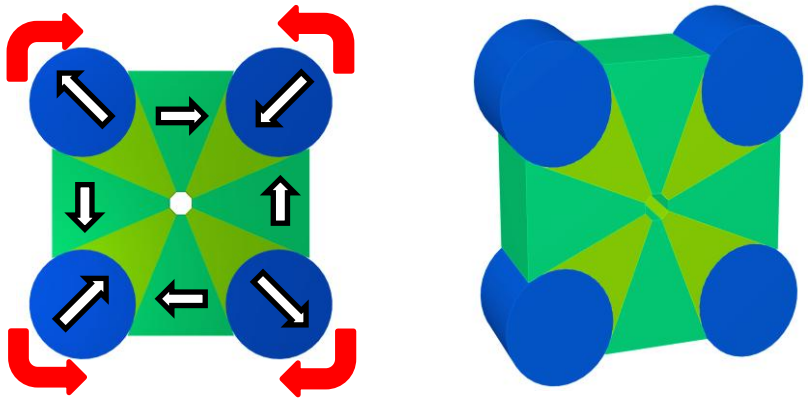
2026 Milestones and Deliverables

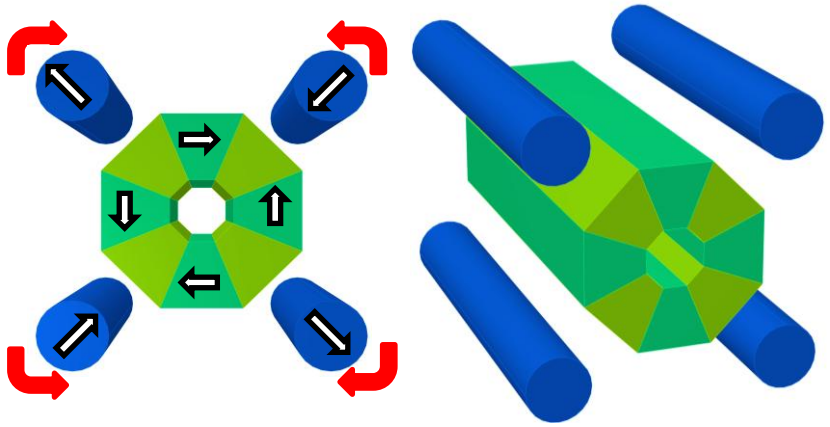
Milestone M1: Magnetic design of compact and normal PMQ [80%]

- Beam Dynamics final check on the EuPRAXIA working point:
  1. A configuration with lower gradient (≈320 T/m) w.r.t. the one proposed for HARMONIQ (495 T/m) with short (Lmag =23 mm) and long (Lmag =46 mm) PMQs
  2. A configuration with higher maximum gradients (≈650 T/m) w.r. t. the one proposed for HARMONIQ (570 T/m) with constant length (Lm=23mm) PMQ
- Configuration 1 has been chosen, furhter study are ongoing to evaluate cross talks between PMQs spaced as well in the Eupraxia layout.
- Preliminary stress analysis on the magnetic blocks have been completed on the stand alone magnet both for compact and normal design.

Delivery M1: Magnetic design of compact and normal PMQ [80%]

- Final magnetic design has to be reached because of several cross-talks analysis are ongoing. Evaluations involves beam dynamics and stress analysis.
- Kick-off Meeting with the company for the production held on the beginning of 2026.
- Design must be freezed and the order must be placed before the end of 2026!





Plan and Financial Request@LNF for 2027

- Magnetic Measurements on the compact" and "normal" PMQs.

Milestones for 2027

- M2: Production of "compact" and "normal" PMQs
- M3: Magnetic measurements of compact and normal PMQ
- M4: Manufacturing of the other PMQ

	Descrizione	Richieste (k€)	Totale Richieste (k€)
consumo	xxx	xxx	xxx
missioni	Sopralluoghi ditte Trasferte dalle altre sezioni per misure magentiche	3	3
inventario			34
	Produzione e assemblaggio PMQ	34	
Totale			**37**

LNF	FTE
A. Vannozzi (Resp. Naz.)	0,5
L. Sabbatini	0,2
A.Trigilio	0,2
S. Pioli	0,1
G. Latini	0,1
A.Giribono	0,1
C. Vaccarezza	0,2
A.Liedl	0,1
R. Pompili	0,1
L. Foggetta	0,2
B. Buonomo	0,2
Tot./Unit	**2,0**

Esperimento in CSN5 (INFN)

SL_BetaTest

(SPARC_LAB Betatron radiation Test)

4 anni (2024-2027)
Units: LNF, RM1, RM2, MI
Research line: Advanced Accelerators

Enrica Chiadroni

(Resp. Naz/Loc.) – LNF

FTE 5.4@LNF

FTE tot 13.9

The main goal of this experiment is the study of emission of betatron radiation from the beam-driven plasma wakefield acceleration at SPARC_LAB, as a test-bed for the development of a plasma-based undulator device. Indeed, conventional undulators are still too cumbersome and expensive to meet the requirements of compactness and sustainability. Nowadays, advanced undulator concepts arouse great interest in pushing the frontier beyond conventional, magnet- based undulators.

2026 Milestones

Milestone M1: Realistic start-to-end simulations: from the setup of electron beam parameters to the generation of betatron radiation [75%]

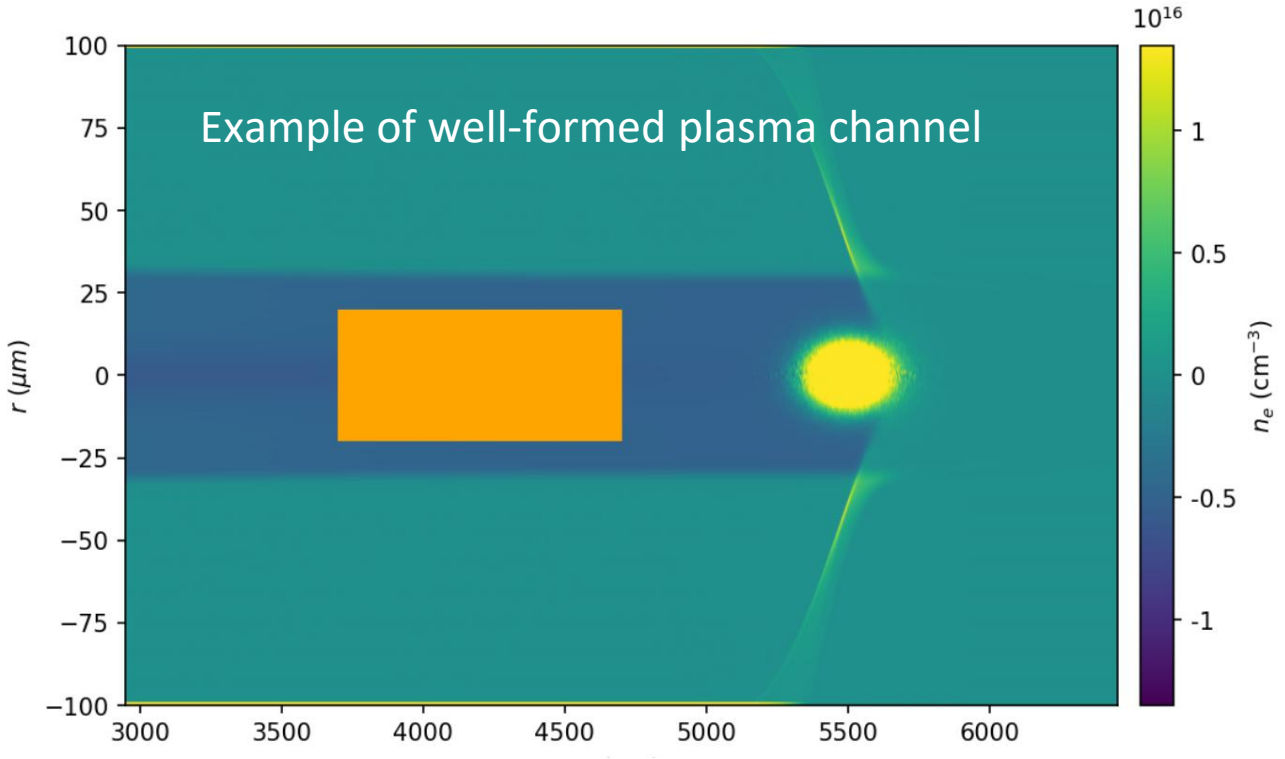
FBPIC simulations to study the plasma channel formation, driven by a particle beam;
ASTRA simulations to study the beam physics for injection into the plasma module.

Milestone M2: Design of plasma diagnostic based on THz radiation [50%]

Definition of the experimental apparatus and acquisition of photoconductive antenna for THz generation and receiver.

Milestone M3: Design of plasma module for betatron emission, i.e., plasma chamber, capillary, extraction electron beamline [50%]

Experimental study of the plasma source.



The parameters are as follows: unperturbed plasma density $n_e = 5 \cdot 10^{15}\,\mathrm{cm}^{-3}$, channel radius $R_{ch} = 30$ µm, bunch charge $Q_b =$ 600 pC, bunch length $\sigma_z = 90$ µm. The orange rectangle sketches the volume employed for measuring the ion column formation.

Plan and Financial Request@LNF for 2027

Experimental set up of plasma diagnostics;
Formation of plasma channel.

Milestones for 2027

Executive design of the experiment from the electron source to the betatron radiation generation [December 2027];

Final demonstration of the feasibility of a plasma undulator based on betatron radiation [December 2027].

		Richieste LNF (k€)	Totale Richieste (k€)
consumo	THz optics (filters and lenses for THz-based plasma diagnostics); capillaries.	20	20
missioni	Partecipazione a workshop e conferenze	3	12
inventario	0	0	15
	Camera for betatron radiation characterization	0	
	0	0	
Totale			**47**

LNF	FTE
Chiadroni (Resp. Naz.)	0.7
Bellaveglia	0.1
Biagioni	0.2
Buitrago	0.8
Costa	0.1
Crincoli	0.4
Curcio	0.5
Del Dotto	0.5
Del Franco	0.2
Demitra	0.4
Ferrario	0.2
Franzini	0.1
Galletti	0.1

LNF	FTE
Gallo	0.1
Giribono	0.1
Piersanti	0.2
Quartullo	0.1
Silvi	0.4
Stella	0.1
Vaccarezza	0.1
Tot./Unit	**5.4**

Invited Talks related to SL_Betatest

- L. Crincoli, Advanced ceramic plasma discharge capillary for high repetition rate operation, European Advanced Accelerator Conference 2025, La Biodola, Italia
- R. Pompili, Guiding of charged particle beams in curved plasma-discharge capillaries, International Particle Accelerator Conference 2025, Taipei, Taiwan
- A. R. Rossi, Ion channel formation for advanced betatron emission, European Advanced Accelerator Conference 2025, La Biodola, Italia

Publications

- A. Curcio et al., Betatron Radiation as a Path to Plasma Undulators: A Case Study at SPARC_LAB, Appl. Sci. 2026, 1, 0 - https://doi.org/10.3390/app1010000
- A. Frazzitta et al., Extreme radiation emission regime for electron beams in strong focusing ion channels and undulators, PRAB
- M. Galletti et al., Femtosecond laser-induced plasma filaments for beam-driven plasma wakefield acceleration, Phys. Rev. E

Thesis

- L. Crincoli, Experimental and theoretical development of m-scale and high repetition rate plasma sources for plasma-based particle accelerators (PhD in Accelerator Physics)
- D. Francescone, Study and characterization of betatron radiation source from plasma wakefield accelerator (PhD in Accelerator Physics)

SuperMAD @ LNF

Istituto Nazionale di Fisica Nucleare

Superconducting **M**aterials for **A**xions **D**etectors

3 anni (2025-2027)
Units: LNF,**LNL**,RM3,TO

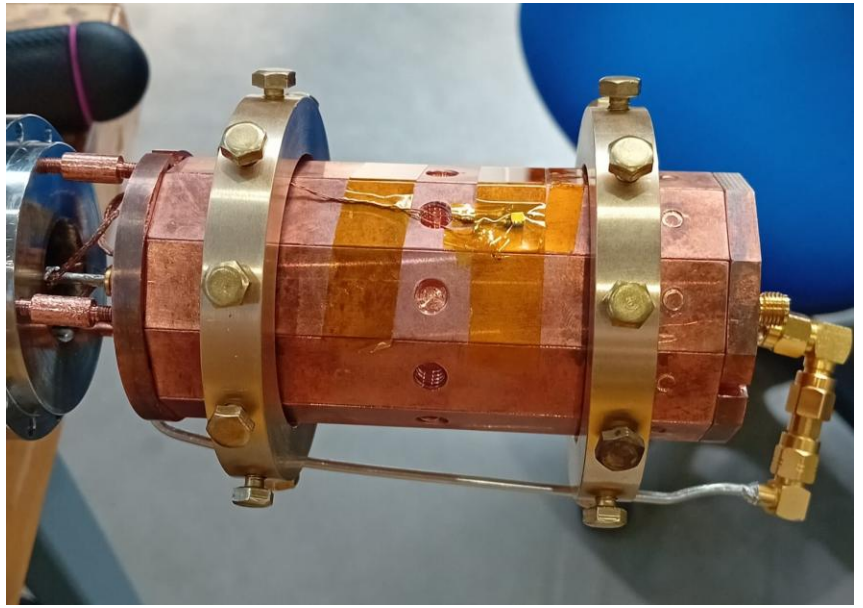
Research line: Accelerator
(for historical reasons being the sequel of SAMARA)

PI (LNL): Cristian Pira

LNF: Giovanni Maccarrone

FTE (2027) 0.5@LNF

FTE (2027) Tot 10.5 (Ric.) + 3.5 (Tec.)



Pic shows the 10-segment reference cavity with the clamping system.
The segments will be replaced with others with the SC internal surface, both HTC and Nb based.

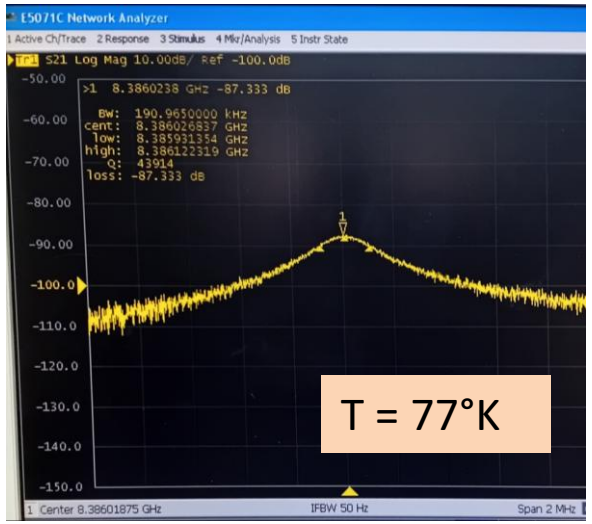
The goal of SuperMAD is to explore, develop and study SC materials to work in the MW regime in high DC magnetic fields, with the aim of producing high-efficiency detectors for the Axions search.

To achieve this, SuperMAD focuses in 3 different aspects:

- Development of SC materials
- Characterization and modeling of vortex dynamics in different materials
- **Design, production and testing of SC cavities (LNF)**

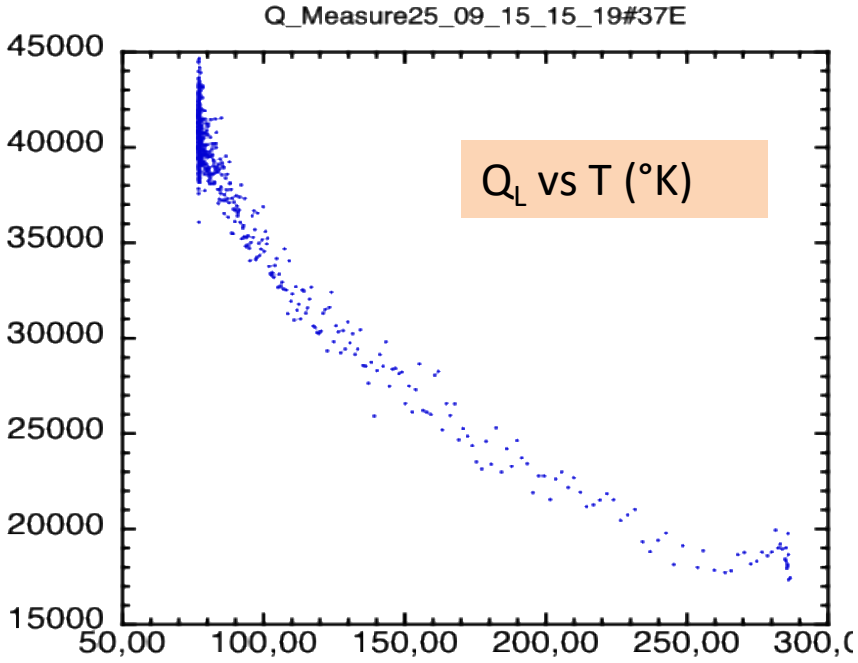
2026 Milestones @ LNF

Milestone M1 (end 2025): Test of the Reference Cavity



VNA transmitted signal:
F peak = 8.386 GHz
Q_L ~ 43900

From Simulation:
F peak = 8.500 GHz
Q ~ 48000





Milestone M2 (end 2026): Test cavity with SC segments



Segment Prep.
Indium-Tin wire



Segment in Owen
T~130°C



Tape soldered.
To be cutted

HTC tape is soldered to a single segment using low-temperature (~120°C) Indium-Tin wire.
Despite this is still an R&D phase, we are confident to have a good cavity to test by the end of the year.

Milestones for 2027

Milestone M3 (end 2027): Test of the final SC Cavity

In the last year of the activity, we want to finalize the design of the SC cavity using both HTC tapes and Nb based segments. Test performed at 4°K.

Milestone M4 (end 2027): Test of Nb3Sn Cavity

This is a different cavity built in LNL using Nb3Sn to be tested in Magnetic field at the lowest temperature.

Milestone M5 (All 3 Years): Feasibility study for FLASH Cavity

Feasibility study for a sputtering cover in Legnaro, using a Nb base SC material (under study) for the Flash prototype cavity built in LNF/Bonn (Flash/Gravenet collaborations) and tested in Frascati.

Requests and FTE for 2027

SuperMAD actiivity moves to “Dotazioni Gruppo V”

GM (local coordinator) will be “in Pensione!” starting from 01/07 next year.
DDG having reached 70 y.o. (in December this year) he will no longer count as an FTE in 2027.

Therefore, while maintaining the scientific commitments, it was decided within the collaboration to transfer the FTE and the financial part of ENEA collaborators to the RomaTre group.

FTE 2027:

Giovanni Maccarrone		30%
Carlo Ligi	10%	
Claudio Gatti (sinergy Gravenet)		10%

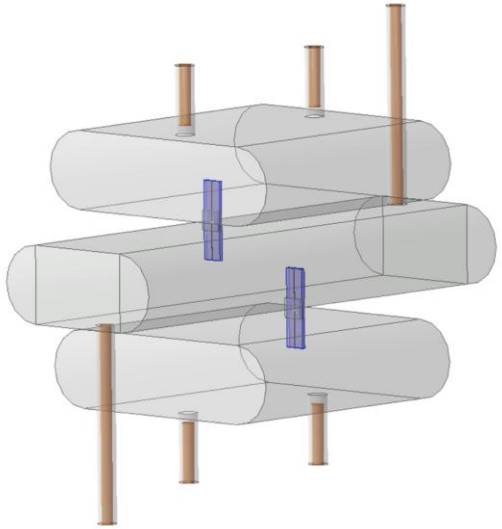
Funds 2027 (KEuro) :

Missioni	1
Consumo (200Liters LHe)	7

Istituto Nazionale di Fisica Nucleare

QUART&T:

***QU**antum **A**rchitectures for **T**heory **& T**echnology*







3 years project (2025-2027)

Research line: **Quantum technologies for fundamental physics**

Resp. Naz.: Andrea GIACHERO (Univ. MiB)

Resp. Loc. LNF: Carlo LIGI

FTE 2.6@LNF

FTE tot 29.1 (2025 proposal)

The QUantum Architectures for Theory & Technology (QUART&T) project aims to develop demonstrator quantum architectures, establishing the foundation for experimental platforms where theoretical models and phenomena of interest to the INFN can be tested.

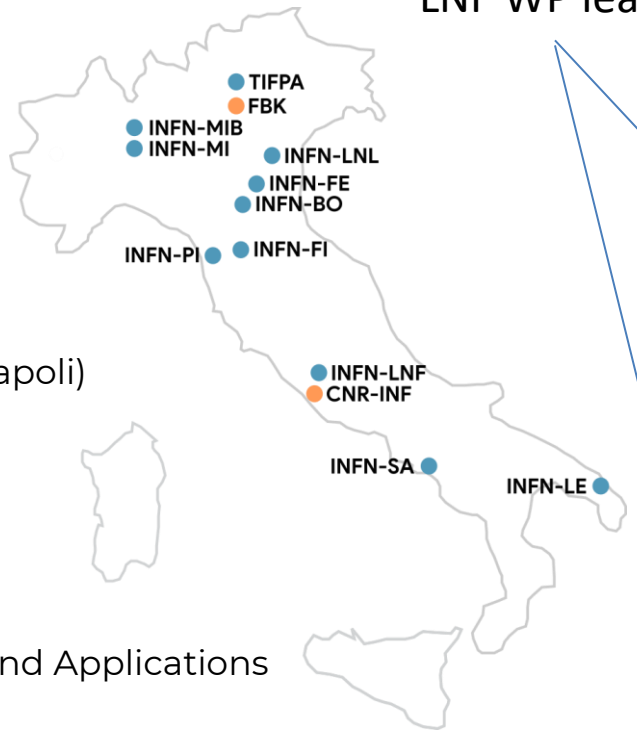
QUART&T: Groups and WPs

8 INFN Units
- INFN Bologna
- INFN Ferrara
- INFN Firenze
- INFN Lecce
- INFN Milano
- INFN Milano Bicocca
- INFN Gruppo Collegato di Salerno (INFN Napoli)

2 INFN National Laboratories
- INFN Laboratori Lazionali di Frascati (LNF)
- INFN Laboratori Lazionali di Legnaro (LNL)

1 INFN Research Center
- Trento Institute for Fundamental Physics and Applications (TIFPA).

2 External Research Centers
- Fondazione Bruno Kessler (FBK, Trento)
- Istituto di Fotonica e Nanotecnologie (CNR-IFN, Roma)



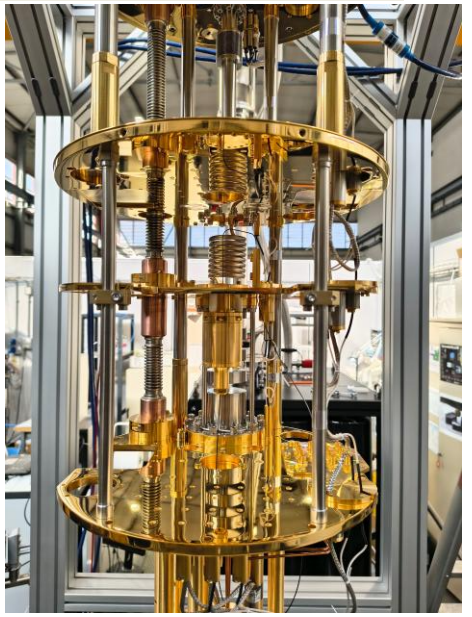
LNF WP leader

- **WP1: Theoretical Background**
  INFN-BO, INFN-FI, INFN-Pi, INFN-FE, INFN-TIFPA
- **WP2: Design and simulations of planar devices**
  INFN-FI, INFN-BO, INFN-MIB, INFN-TIFPA
- **WP3: Design and simulations of 3D devices**
  INFN-LNF WP leader (A D'Elia), INFN-LE, INFN-LNL
- **WP4: Qubit Fabrication**
  INFN-LNF, INFN-LNL, INFN-TIFPA, FBK, and CNR-IFN
- **WP5: Software and readout development**
  INFN-BO, INFN-FE, INFN-MI, INFN-MIB, **INFN-LNF**
- **WP6: Quantum limited amplification**
  INFN-MIB, INFN-LNF, INFN-SA, INFN-TIFPA, FBK
- **WP7: Packing optimization**
  INFN-LNF WP leader (A Rettaroli), INFN-TIFPA, INFN-PI
- **WP8: Characterization, Measurement and Simulation demonstration**
  INFN-LNF, INFN-BO,INFN-LE, INFN-MIB, INFN-PI, INFN-TIFPA
- **WP9: Application for quantum sensing and ML**
  INFN-LNF, INFN-LE, INFN-MIB, INFN-MI, INFN-PI
- **WP10: Management and Communication**
  INFN-MIB and all the other involved units

QUART&T: Attività a LNF 2026 (1)

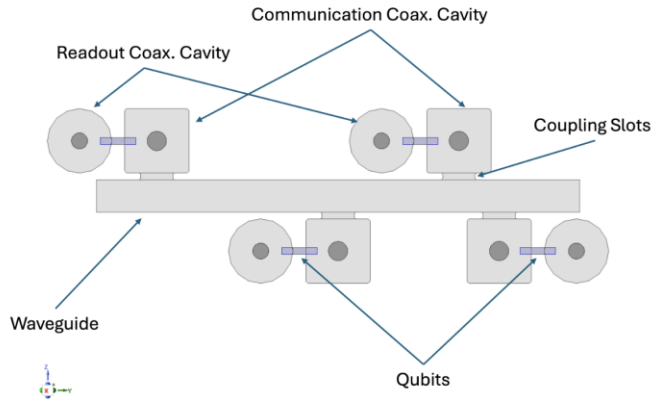
- Characterization of a new cryostat with a dilution refrigerator dedicated to quantum devices





- Characterization of a triple cavity for multi-qub photon detection schemes



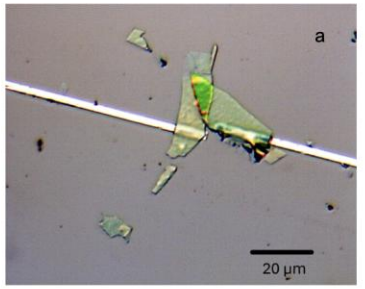


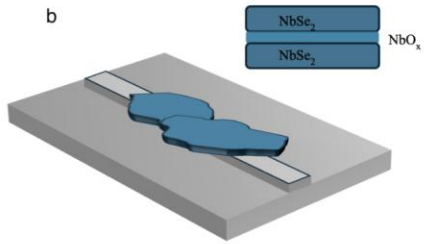
- Electromagnetic simulation of 3D quantum router for all-to-all connectivity

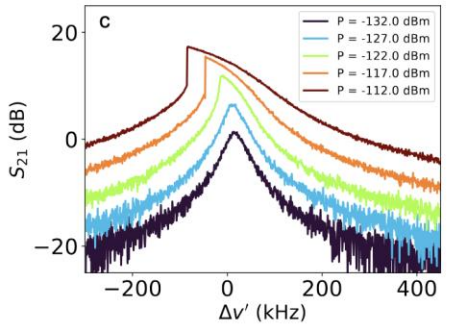
QUART&T: Attività a LNF 2026 (2)

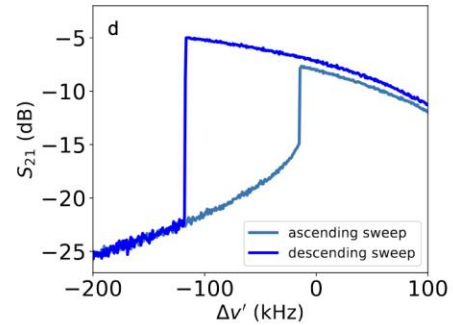
- Ongoing characterization of transmon qubit in a single Al cavity
- Successful fabrication+characterization of the FIRST NbSe2 qubit (!)
- Multi qubit simulation for quantum sensing and machine learning

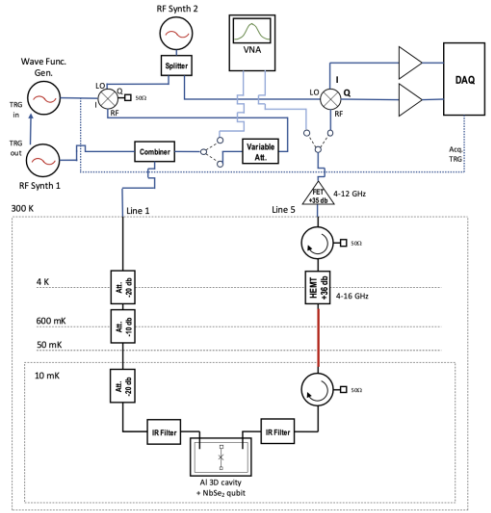
Ongoing purchase procedure for qubits from CON-SCIENCE for the triple cavity dedicated to quantum sensing tests using quantum machine learning protocols (collaboration with Rome Univ. – S. Giagu)











QUART&T: 2027

Richieste LNF:

20 k€ fabbricazione dispositivi

10 k€ motore criogenico

15 k€ consumi + barra Allum. + barra zaffiro + compressore per refrigeratore

5 k€ missioni

Attività 2027:

- Test Quantum Amplifiers (TWPA)
- Test with single cavity with Qubit coupled to a TWPA for quantum sensing + machine learning
- Test for 3D cavity tuning with a sapphire rod using a cryogenic motor
- A new master student will work on qubit optimal control

Milestones 2027:

M7.3: Report on the array of coupled devices, both 2D and 3D (M36)

M8.3: Application of the multi-qubit device for machine learning (M36)

FTE LNF 2027	
C. Gatti	0% (+20% GRAVNET)
C. Ligi (RL)	5% (+25% GRAVNET)
A. D'Elia	0% (+50% GRAVNET)
M. Beretta	10% (+15% GRAVNET)
L. Piersanti	10% (+15% GRAVNET)
A. Rettaroli	0% (+20% GRAVNET)
S. Tocci	0% (+20% GRAVNET)
B. Buonomo	20%
F. Chiarello (CNR)	30%
F. Mattioli (CNR)	20%
TOT LNF	**2.6 FTE (0.95 + 1.65 GRAVNET)**

Publications

- **Transmon Qubit Modeling and Characterization for Dark Matter Search** *(R. Moretti et al. – IEEE Trans., 2026)*
- **Enhancing axion searches with quantum coincidence in superconducting qubits** *(A. S. Piedjou Komnang et al. – Il Nuovo Cimento C, 2026)*
- **Coherent oscillations in weakly anharmonic NbSe2 qubit** *(A. D'Elia – submitted)*

Thanks for you attention

I WOULD LIKE TO THANK EVERYONE WHO HAS SUBMITTED MATERIAL FOR THIS PRESENTATION AND CONTRIBUTES TO CSN5 AT THE LNF!



Procedura di Ranking dei Proposal di Esperimenti Standard

- Ad ogni nuovo proposal di esperimento standard è assegnato un ranking numerico (n $\in \mathbb{N}$) suddiviso in 3 voci.
- Impatto scientifico/tecnico. Chiarezza nella spiegazione della novelty e sull'incremento rispetto allo stato dell'arte. Attinenza con le mission dell'Ente. Attinenza con le tematiche della CSN5. (1-10)
- Chiarezza nella spiegazione delle metodologie. Chiarezza nella spiegazione dei risultati attesi. Metodi di valutazione del progresso dell'esperimento. Analisi e mitigazione dei rischi. (1-10)
- Sostenibilità dell'esperimento. Consistenza e giustificazione delle richieste finanziarie. Consistenza del personale impegnato nell'esperimento con l'attività prevista. Competenza dei proponenti riguardo l'argomento della proposta. (1-10)
- Il voto di ogni voce verrà accompagnato da un giudizio.

Giudizi e Selezione Finale

- I voti vanno accompagnati da giudizi puntuali e dettagliati dai quali si evincono le ragioni del voto.
- I giudizi vanno presi anche come indicazioni utili a migliorare la proposta.
- Ogni sottocommissione stila i giudizi collegialmente preoccupandosi della coerenza fra termini e/o aggettivi con il risultato finale.
- Per passare la prima fase un proposal deve totalizzare almeno 18/30 **e (**v**)** deve aver preso un voto maggiore di 4/10 su tutte le voci.
- Gli esperimenti respinti potranno ritentare l'anno successivo tenendo presenti le osservazioni della Commissione.

BOND - CRYSTAL BOND

Progetto CALL CSNV 2025-2027

Acronimo e Titolo: BOND - Crystal BOND
Area di ricerca: Acceleratori
Responsabile nazionale: Andrea Mazzolari
Budget 2026 (LNF): 22keuro

Unità partecipanti:
Ferrara
Legnaro (D. De Salvador)
Trieste (F. Arfelli)
Frascati (S. Dabagov) (1.05 FTE)

Supporto e scopo

The unifying goal is to establish INFN as a global leader in the fabrication of crystals for radiation manipulation.



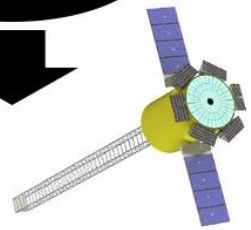
CERN



Sincrotroni:
ESRF (Francia)
XFEL (Germania)
Lund University (Svezia)
ELETTRA (Italia)
University of Saskatchewan (Canada)

BOND technology

CINEL (Italia) – Industria



Astrofisica
INAF (Italia)
Università degli studi di Trieste (Italia)

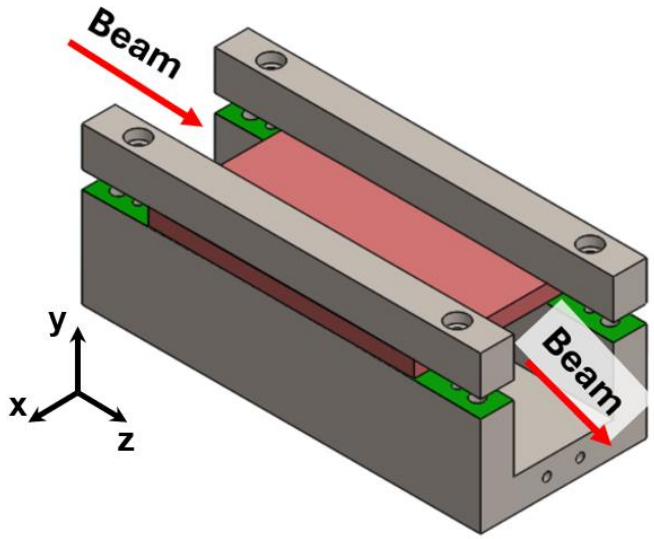
Deflessione di fasci di particelle @ CERN

Deflessione di fasci di particelle a «largo angolo». Di interesse per CERN-LHC e futuri acceleratori: tecnologia abilitante per esperimenti a targetta fissa

Tecnologia attuale

Deformazione del cristallo imposta da supporti meccanici.

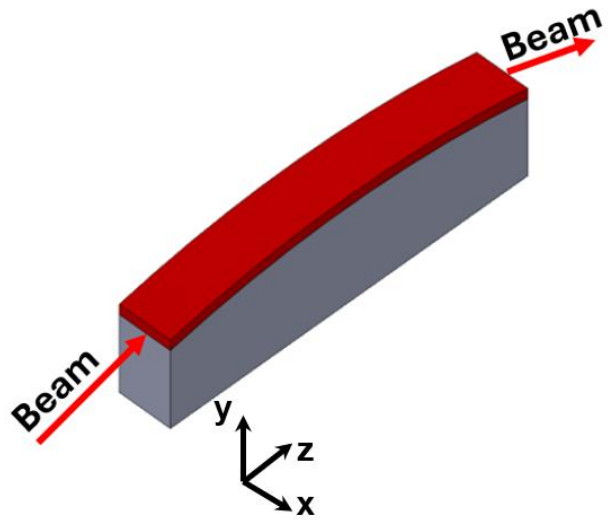
- Ingombranti
- Pesanti
- Difficili da realizzare perché richieste tolleranze stringenti



Tecnologia proposta

Cristallo in silicio deformato tramite «anodic bonding» di silicio su vetro

- Assemblato leggero
- Ingombro ridotto
- Curvatura estremamente uniforme

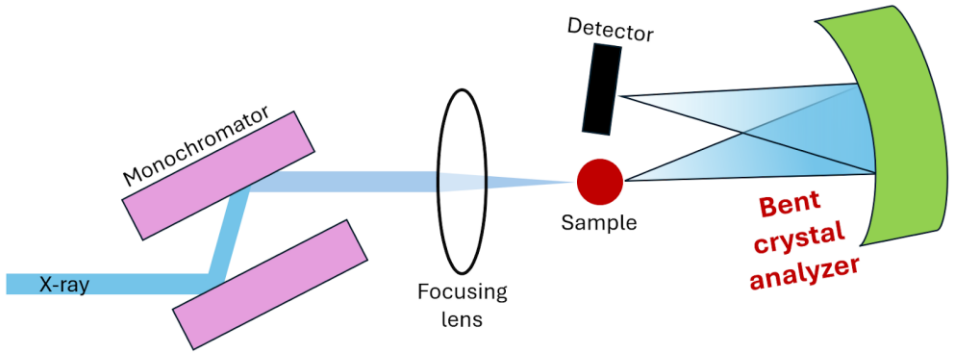


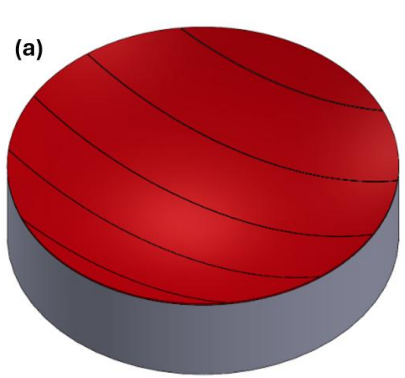
Ottiche per raggi-x @ sinctrotroni

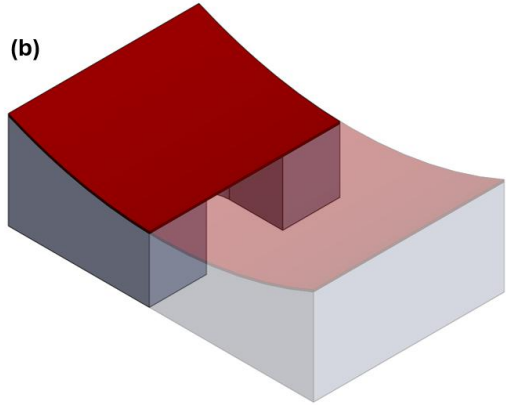
Sviluppo di innovative ottiche per raggi-x da impiegare per tecniche di imaging presso sincrotroni.

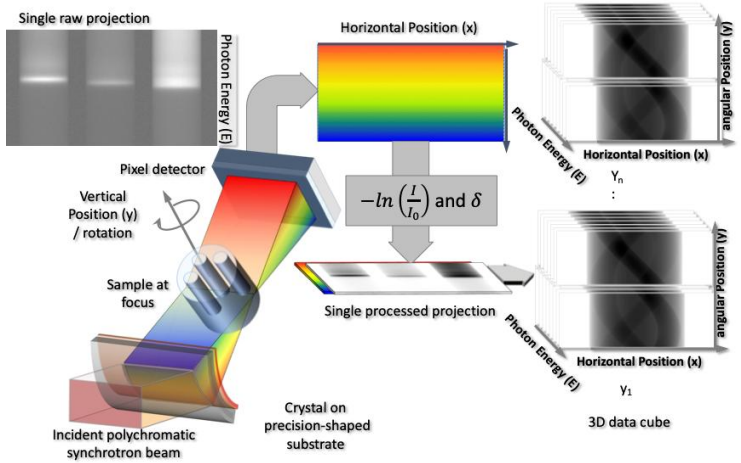
Materiali innovativi quali germanio, quarzo, niobato di litio.

Cristalli deformati tramite anodic bonding e thermo-compression bonding.

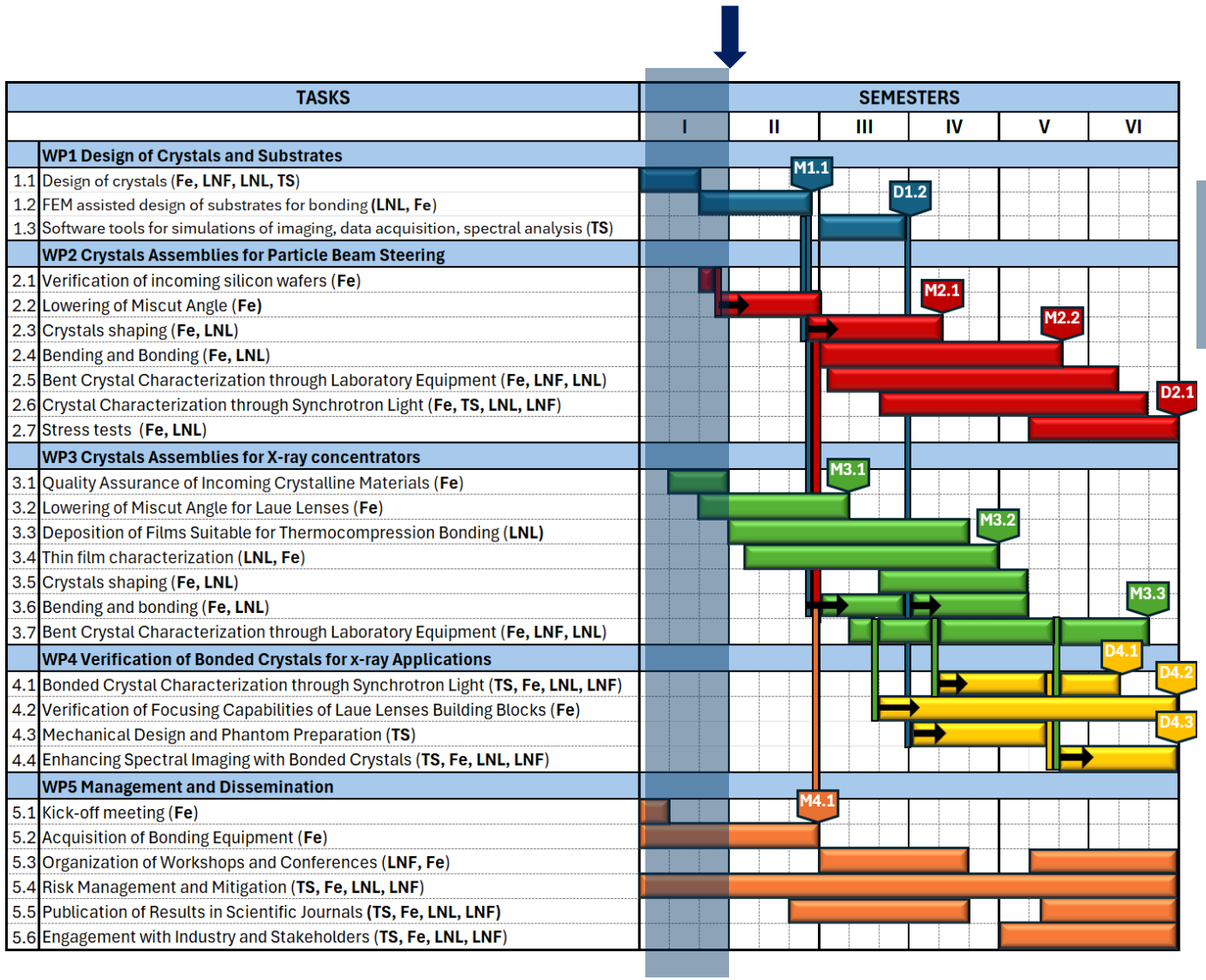








Cronoprogramma



Attività distribuita in 3 anni

- Primo anno progettazione (LNF) ed acquisizione strumentazione
  - primi passi di progettazione ottica/cristalli

- Secondo anno R&D

- Terzo anno R&D e testing.